Points of Interest as Crime Generators: A Supervised Regression Analysis of London Ward-Level Crime Density

> *I, Emily Deeb, pledge my honour that the work presented in this assessment is my own. Where information has been derived from other sources, I confirm that this has been indicated in the work.*

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Preparation

- Github link
- **Number of words:** 1,500 words
- **Runtime:** ~1 minute (pre-processed data loaded from file; full data extraction and model training estimated ~45–60 minutes on first run)

Coding environment: Miniconda 3 · Python 3.x · urbsim environment
Hardware: Intel i7-10700 @ 2.90GHz · 10GB RAM
Key packages: geopandas, scikit-learn, shap, osmnx, matplotlib, plotly

In [57]:
```
import time
start_time = time.time()
```

In [58]:
```
# Step 01: Imports + BASE URL
import pandas as pd
import geopandas as gpd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.colors as mcolors
import seaborn as sns
import osmnx as ox
import tempfile, os, requests

BASE = "https://raw.githubusercontent.com/EmilyDeeb/0006_Assessment/main/data/"
```

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Introduction

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This study examines the spatial relationship between urban amenity density and crime concentration across London wards. We use open-source point-of-interest (POI) data from OpenStreetMap and recorded crime data from the Metropolitan Police Service within a supervised regression model to assess whether the spatial layout of urban amenities such as cafés, clubs, and restaurants predicts crime density and whether specific amenity types are associated with different crime categories.

Routine Activity Theory (Cohen and Felson, 1979) argues that crime occurs when motivated offenders, suitable targets, and the absence of capable guardians intersect in the same place and time. This convergence can be defined by the functional land-use profile of an area; the mix of amenities, services, and commercial activity that determines who is present and when. Brantingham and Brantingham (1995) formalise this with the concept of crime generators, theorising that locations that attract large numbers of people for non-criminal reasons increase crime opportunities by providing many possible offenders and targets.

POI data has become a key resource in urban crime research, with Cichosz (2020) demonstrating that OSM POI density features outperform traditional land-use classifications, and Andresen (2011) arguing that ambient population — proxied here by POI density, is a more meaningful crime denominator than residential population.

Although significant progress has been made in analysing the relationship between crime and urban fabric, most studies model total crime or individual crime types in isolation, mixing together opportunity structures that operate differently by crime type, a distinction this study is designed to test. To address this gap, we split the crime dataset into two theoretically motivated categories; Type A (person/society crime) and Type B (property crime), and apply SHAP decomposition to compare which amenity types drive each, enabling a more granular account of how urban land use shapes different forms of crime across London wards. The following research questions frame the analysis.

Research question

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Primary: To what extent does the density of urban amenities predict person/society and property crime densities across London wards?

Sub-question: Which specific amenity categories serve as the primary crime generators for each crime type, and do these differ systematically between person/society and property crime?

Data and Feature Engineering

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Three datasets were assembled to address these questions.

Dataset 01: Metropolitan Police Service recorded crime data were obtained from the London Datastore (data.london.gov.uk), containing monthly crime counts by ward and crime category for the 24 months to February 2025. Crimes were summed across the full period and aggregated into two theoretically motivated categories, as seen in Table 1:

Table 1: Crime category definitions and offence types included in the analysis

Crime Category	Type	Offence Types Included	Total Crimes Recorded
Type A — Person/Society Crime	Target variable	Violence against the person, robbery, sexual offences, possession of weapons, drug offences, public order offences, miscellaneous crimes against society, fraude and forgery	813,720 crimes
Type B — Property Crime	Target variable	Theft, burglary, vehicle offences, arson and criminal damage	968,439 crimes

Note: *NFIB fraud offences were excluded from both categories as fraud is recorded at the victim's home address rather than the location of the offence, rendering it spatially uninformative at the ward level. Type B (property crime) exceeds Type A (person/society crime) by approximately 19%.*

In [59]:
```
# Data 01: Crime data
# Load crime → aggregate Type A + Type B → crime densities
# Sum monthly crime counts across all 12 months to get annual crime counts,
# then divide by area to get crime densities.
# ==============================================================================
# Plot Figure 01: Total recorded crimes by crime type (Type A vs Type B) → bar chart
```

In [60]:
```
# Load crime → aggregate Type A + Type B → crime densities
crime_raw = pd.read_csv(BASE + "crime.csv")

# Type A — Person/Society crime categories
typeA_cats = [
    'VIOLENCE AGAINST THE PERSON', 'ROBBERY',
    'SEXUAL OFFENCES', 'POSSESSION OF WEAPONS',
    'DRUG OFFENCES', 'PUBLIC ORDER OFFENCES',
    'MISCELLANEOUS CRIMES AGAINST SOCIETY', 'FRAUD AND FORGERY'
]

# Type B — Property crime
typeB_cats = [
    'THEFT', 'BURGLARY',
    'VEHICLE OFFENCES', 'ARSON AND CRIMINAL DAMAGE'
]
```

In [61]:
```
# Sum monthly columns to annual count
month_cols = [c for c in crime_raw.columns if str(c).startswith("2")]
crime_raw["annual_count"] = crime_raw[month_cols].sum(axis=1)

def aggregate_crime(df, categories, label):
    """Filter to crime categories and sum to ward level."""
    return (
        df[df["MajorText"].isin(categories)]
        .groupby(["WardCode", "WardName", "LookUp_BoroughName"])["annual_count"]
        .sum()
        .reset_index()
        .rename(columns={
            "annual_count": label,
            "LookUp_BoroughName": "Borough"
        })
    )

crime_A = aggregate_crime(crime_raw, typeA_cats, "typeA_count")
crime_B = aggregate_crime(crime_raw, typeB_cats, "typeB_count")

ward_crime = crime_A.merge(
    crime_B[["WardCode", "typeB_count"]], on="WardCode", how="left"
).fillna(0)

print(f"Wards: {len(ward_crime)}")
ward_crime.head()
```

```
Wards: 679
```

Out[61]:

	WardCode	WardName	Borough	typeA_count	typeB_count
0	E05009317	Bethnal Green	Tower Hamlets	2811	2403
1	E05009318	Blackwall & Cubitt Town	Tower Hamlets	1776	1615
2	E05009319	Bow East	Tower Hamlets	2085	1973
3	E05009320	Bow West	Tower Hamlets	1422	1457
4	E05009321	Bromley North	Tower Hamlets	1482	1403

Dataset 02: Ward Spatial Boundary and population as shown in Table 2 **Table 2:** Spatial and demographic data sources

Dataset	Source	URL	Geography	Notes
Ward boundary polygons	ONS Open Geography Portal	geoportal.statistics.gov.uk	Ward level	2024 boundaries, filtered to London using GLA boundary
Ward areas	Derived from boundary polygons	—	Ward level	Calculated using British National Grid projection (EPSG:27700)
Population estimates	ONS mid-2021 ward-level estimates	geoportal.statistics.gov.uk	Ward level	No missing values after merging; 679 London wards retained

Note: Population density was retained in the dataset but excluded from model features to isolate the contribution of amenity configuration independently of population size. Incorporating it as a control variable is a natural next step.

In [62]:

```
# Data 02: Ward Spatial Data
# load population and area data
# load GLA boundary + filter 2024 wards to London
# Centroid of each ward
# spatial join to GLA boundary to filter to London wards
# calculate area in km2
# Build main GeoDataFrame -> compute crime densities using ward geometries + crime counts
# ==============================================================================
# Table 3: Crime density descriptive stats
```

In [63]:

```
# Load population
pop = pd.read_csv(BASE + "ward_population.csv", encoding="latin-1")

# Rename columns and select relevant ones
pop = pop.rename(columns={
    "Ward 2023 Code": "WardCode",
    "Total": "population"
})[["WardCode", "population"]]

# Remove commas and convert population to int
pop["population"] = (
    pop["population"].astype(str)
    .str.replace(",", "")
    .astype(int)
)

# Filter to E05 wards only (Greater London)
pop = pop[pop["WardCode"].str.startswith("E05")].copy()

# Merge population data with crime data
ward_crime = ward_crime.merge(pop, on="WardCode", how="left")

# Check for missing population values after merge
print(f"Missing population: {ward_crime['population'].isna().sum()}")
```

```
Missing population: 0
```

In [64]:

```
# Spatial join to filter to wards whose centroids fall within GLA boundary
london_wards = gpd.sjoin(
    wards_centroids,
    gla_gdf[["geometry"]],
    how="inner", predicate="within"
)

# Calculate area in km2 using projected CRS for accurate area calculation
london_wards = wards_2024.loc[london_wards.index].copy()
london_wards = london_wards.rename(columns={
    "WD24CD": "WardCode",
    "WD24NM": "WardName_geo"
})

# Remove non-geographic wards
london_wards = london_wards[
    ~london_wards["WardName_geo"].str.contains(
```

```
        'Aviation|SO18', case=False, na=False
    )
].copy()

# Calculate area in km2 using projected CRS for accurate area calculation
london_wards["area_km2"] = (
    london_wards.to_crs("EPSG:27700").geometry.area / 1_000_000
)

# Check final wards + area
london_wards[['WardCode', 'WardName_geo', 'area_km2']].head()
```

Out[64]:

	WardCode	WardName_geo	area_km2
1104	E05009288	Aldersgate	0.098405
1105	E05009289	Aldgate	0.114784
1106	E05009290	Bassishaw	0.106979
1107	E05009291	Billingsgate	0.086042
1108	E05009292	Bishopsgate	0.254265

In [65]:

```
print(f"Ward count after filter: {len(london_wards)}")
```

```
Ward count after filter: 704
```

In [66]:

```
# Note: 25 wards in the 2024 boundary file have no matching crime records
# (E05009288–E05009312) — likely new ward codes created in the 2024 boundary
# revision that postdate the MPS crime dataset. These are dropped automatically
# in Step 05 via the left merge and dropna on geometry/area_km2.
# Final analysis uses 679 matched wards.
print(f"Ward count after filter: {len(london_wards)}")
print("Note: 25 unmatched wards will be dropped at merge stage — expected behaviour")
```

```
Ward count after filter: 704
Note: 25 unmatched wards will be dropped at merge stage — expected behaviour
```

In [67]:

```
# Build main GeoDataFrame
# Merge crime counts with ward geometries
ward_gdf = ward_crime.merge(
    london_wards[["WardCode", "area_km2", "geometry"]],
    on="WardCode",
    how="left",
)

# Compute crime density (target variables)
ward_gdf["typeA_density"] = ward_gdf["typeA_count"] / ward_gdf["area_km2"]
ward_gdf["typeB_density"] = ward_gdf["typeB_count"] / ward_gdf["area_km2"]

# Promote to GeoDataFrame and drop rows with missing spatial data
ward_gdf = (
    gpd.GeoDataFrame(ward_gdf, geometry="geometry", crs="EPSG:4326")
    .dropna(subset=["geometry", "area_km2"])
)

print(f"Final ward count:    {len(ward_gdf)}")
print(f"Missing geometries:  {ward_gdf['geometry'].isna().sum()}")

# ── Styled summary: crime density descriptive stats (pre-transformation) ────

STYLE = {
    "font-family": "Inter, sans-serif",
    "font-size":   "12px",
    "color":       "#1B2632",
    "border-bottom": "1px solid #C9C1B1",
    "padding":     "7px 12px",
}

TABLE_STYLES = [
    {
        "selector": "thead tr",
        "props": [
            ("background-color", "#2C3B4D"),
            ("color",            "#EEE9DF"),
            ("font-weight",      "500"),
        ],
    },
    {
        "selector": "caption",
        "props": [
            ("font-size",   "11px"),
            ("color",       "#C9C1B1"),
            ("font-style",  "italic"),
            ("text-align",  "left"),
            ("padding",     "6px 0"),
        ],
    },
]

(
    ward_gdf[["typeA_density", "typeB_density"]]
    .describe()
    .round(2)
```

```
    .style
    .format("{:.2f}")
    .set_properties(**STYLE)
    .set_table_styles(TABLE_STYLES)
)
```

```
Final ward count:    679
Missing geometries:  0
```

Out[67]:

	typeA_density	typeB_density
count	679.00	679.00
mean	868.35	1053.34
std	765.14	1505.26
min	8.47	8.56
25%	322.69	304.79
50%	619.01	596.75
75%	1199.32	1237.23
max	4978.44	24964.44

Table 3: Crime density descriptive statistics (crimes per km²) — pre-feature engineering. *Raw crime density is unsuitable for direct modelling: both variables are strongly right-skewed, with standard deviations above their means (* `typeA_density` *: σ=765, μ=868;* `typeB_density` *: σ=1505, μ=1053). Crime concentrates in a few high-density wards, inflating upper values, especially for property crime. A log transformation is applied next to address this.*

Dataset 03: Point-of-interest data were extracted from OpenStreetMap using the OSMnx Python library (Boeing, 2017). After filtering to 17 predefined categories selected as proxies for footfall and economic activity, 41,502 POIs were retained. The Table 4 below shows each category, its OSM tag, and total count.

In [68]:

```
# Data 03: Point of Interest (POI) Data
# Download POIs from OpenStreetMap (points and polygons centroids)
# Clean Point data → filter noise → spatial join to GLA boundary → compute densities
# ================================================================================
# Table 4: POI categories, OSM tags, and counts
# Table 5: POI density descriptive stats
```

In [69]:

```
# Download POIs from OpenStreetMap
gla_polygon = gla_gdf.to_crs("EPSG:4326").union_all()

tags = {
    "amenity": ["pub", "bar", "nightclub", "fast_food", "restaurant",
                "cafe", "bank", "atm", "pharmacy"],
    "railway": ["station"],
    "shop": ["supermarket", "convenience", "mall", "department_store",
             "clothes", "electronics", "jewelry"]
}

print("Downloading POIs — this takes 5-10 minutes...")
poi_raw = ox.features.features_from_polygon(gla_polygon, tags=tags) #, which returns a GeoDataFrame of all n

# Include both points AND polygon centroids
# Many venues (pubs, restaurants) are mapped as building polygons in OSM
poi_points = poi_raw[poi_raw.geometry.type == "Point"].copy()
poi_polys  = poi_raw[poi_raw.geometry.geom_type.isin(["Polygon", "MultiPolygon"])].copy()
poi_polys["geometry"] = poi_polys.to_crs("EPSG:27700").geometry.centroid.to_crs("EPSG:4326")

poi_gdf = pd.concat([poi_points, poi_polys], ignore_index=True)

# Extract poi_type — check amenity, shop, railway in priority order
poi_gdf["poi_type"] = (
    poi_gdf["amenity"]
    .fillna(poi_gdf.get("shop", pd.NA))
    .fillna(poi_gdf.get("railway", pd.NA))
)

poi_gdf = poi_gdf[["geometry", "poi_type"]].dropna()
poi_gdf = poi_gdf.set_crs("EPSG:4326")

print(f"Total POIs (points + polygon centroids): {len(poi_gdf)}")
# print(poi_gdf["poi_type"].value_counts())
# We have extra points because some venues are mapped as both points and polygons in OSM. We keep both to co
# remove doctors
```

```
Downloading POIs — this takes 5-10 minutes...
Total POIs (points + polygon centroids): 41523
```

In [70]:

```
# Filter noise → spatial join → densities
valid_types = [
    "pub", "bar", "nightclub", "fast_food", "restaurant", "cafe",
    "bank", "atm", "pharmacy", "station",
    "supermarket", "convenience", "mall", "department_store",
    "clothes", "electronics", "jewelry"
]
```

```python
# Filter to valid POI types only — this removes ~1,000 noisy points (mostly doctors, which are often mapped
poi_gdf = poi_gdf[poi_gdf["poi_type"].isin(valid_types)].copy()
print(f"POIs after cleaning: {len(poi_gdf)}")

# Spatial join to count POIs within each ward
wards_4326 = ward_gdf[["WardCode", "geometry"]].to_crs("EPSG:4326")
poi_in_wards = gpd.sjoin(poi_gdf, wards_4326, how="left", predicate="within")

poi_counts = (
    poi_in_wards
    .groupby(["WardCode", "poi_type"])
    .size()
    .unstack(fill_value=0)
    .reset_index()
)
poi_counts.columns.name = None

# Drop existing POI/density columns if re-running
existing = [c for c in poi_counts.columns if c != "WardCode" and c in ward_gdf.columns]
existing += [c for c in ward_gdf.columns if c.endswith("_density")
             and c not in ["typeA_density", "typeB_density"]]
ward_gdf = ward_gdf.drop(columns=existing, errors="ignore")

# Merge — left join preserves all 679 wards
ward_gdf = ward_gdf.merge(poi_counts, on="WardCode", how="left")

# Ensure columns exist for wards with no POIs
poi_cols = [c for c in poi_counts.columns if c != "WardCode"]
for col in poi_cols:
    if col not in ward_gdf.columns:
        ward_gdf[col] = 0
ward_gdf[poi_cols] = ward_gdf[poi_cols].fillna(0)

# Compute densities
for col in poi_cols:
    ward_gdf[f"{col}_density"] = ward_gdf[col] / ward_gdf["area_km2"]

poi_density_cols = [f"{c}_density" for c in poi_cols]

# Report empty wards
empty = ward_gdf[ward_gdf[poi_cols].sum(axis=1) == 0][["WardCode", "WardName", "Borough"]]
print(f"Wards with no POIs: {len(empty)}")
print(empty.to_string(index=False))
print(f"\nPOI density features: {len(poi_density_cols)}")
```

```
POIs after cleaning: 41502
Wards with no POIs: 0
Empty DataFrame
Columns: [WardCode, WardName, Borough]
Index: []

POI density features: 17
```

In [71]:

```python
# Table : POI categories, OSM tags, and counts
tag_map = {
    'restaurant':       ('Restaurant',        'amenity=restaurant'),
    'cafe':             ('Cafe',              'amenity=cafe'),
    'fast_food':        ('Fast food',         'amenity=fast_food'),
    'convenience':      ('Convenience store', 'shop=convenience'),
    'clothes':          ('Clothes shop',      'shop=clothes'),
    'atm':              ('ATM',               'amenity=atm'),
    'pub':              ('Pub',               'amenity=pub'),
    'pharmacy':         ('Pharmacy',          'amenity=pharmacy'),
    'bar':              ('Bar',               'amenity=bar'),
    'supermarket':      ('Supermarket',       'shop=supermarket'),
    'station':          ('Rail station',      'railway=station'),
    'bank':             ('Bank',              'amenity=bank'),
    'jewelry':          ('Jewellery shop',    'shop=jewelry'),
    'electronics':      ('Electronics shop',  'shop=electronics'),
    'nightclub':        ('Nightclub',         'amenity=nightclub'),
    'department_store': ('Department store',  'shop=department_store'),
    'mall':             ('Mall',              'shop=mall'),
}

counts = poi_gdf["poi_type"].value_counts()

rows = []
for poi_type, (label, tag) in tag_map.items():
    count = counts.get(poi_type, 0)
    rows.append({'POI Category': label, 'OSM Tag': tag, 'Count': int(count)})

table3 = pd.DataFrame(rows).sort_values('Count', ascending=False).reset_index(drop=True)

total_row = pd.DataFrame([{
    'POI Category': 'Total',
    'OSM Tag': '',
    'Count': table3['Count'].sum()
}])
table3 = pd.concat([table3, total_row], ignore_index=True)

(table3.style
    .format({'Count': '{:,.0f}'})
    .set_properties(**{
```

```
        'font-family': 'Inter, sans-serif',
        'font-size': '13px',
        'text-align': 'left',
        'color': '#1B2632',
        'background-color': '#F7F4ED',
        'border-bottom': '1px solid #C9C1B1',
        'padding': '8px 14px'
    })
    .set_table_styles([
        {'selector': 'thead tr',
         'props': [('background-color', '#2C3B4D'),
                   ('color', '#EEE9DF'),
                   ('font-weight', '500'),
                   ('font-size', '13px')]},
        {'selector': 'tbody tr:nth-child(even)',
         'props': [('background-color', '#EAE6DE')]},
        {'selector': 'tr:last-child',
         'props': [('font-weight', 'bold'),
                   ('border-top', '2px solid #2C3B4D')]},
        {'selector': 'caption',
         'props': [('font-size', '11px'),
                   ('color', '#C9C1B1'),
                   ('font-style', 'italic'),
                   ('text-align', 'left'),
                   ('padding', '6px 0')]}
    ])
    .hide(axis='index')
)
```

Out[71]:

POI Category	OSM Tag	Count
Restaurant	amenity=restaurant	8,978
Cafe	amenity=cafe	6,718
Fast food	amenity=fast_food	6,158
Convenience store	shop=convenience	5,432
Pub	amenity=pub	3,177
Clothes shop	shop=clothes	2,933
ATM	amenity=atm	1,674
Pharmacy	amenity=pharmacy	1,280
Supermarket	shop=supermarket	1,252
Bar	amenity=bar	1,215
Bank	amenity=bank	703
Jewellery shop	shop=jewelry	695
Rail station	railway=station	616
Electronics shop	shop=electronics	300
Nightclub	amenity=nightclub	154
Department store	shop=department_store	134
Mall	shop=mall	83
Total		**41,502**

Table 4: Points of Interest (POI) Extracted from OpenStreetMap — Greater London. *Food and beverage venues dominate — restaurants, cafés, fast food, pubs, and bars account for 26,246 POIs (63% of total).*

In [72]:

```
# POI density descriptive statistics
from matplotlib.colors import LinearSegmentedColormap

# Custom palette for gradient
palette_cmap = LinearSegmentedColormap.from_list(
    'palette', ['#F7F4ED', '#C9C1B1', "#A8696F"]
)

# Build describe table
desc = ward_gdf[poi_density_cols].describe().round(3).T

# Replace count with total POI count per category
desc['count'] = [ward_gdf[col.replace('_density', '')].sum()
                 for col in poi_density_cols]

(desc
    .drop(columns=['min'])
    .rename(columns={'count': 'Total POIs', 'mean': 'Mean', 'std': 'Std',
                     '25%': 'Q1', '50%': 'Median', '75%': 'Q3', 'max': 'Max'})
    .style
    .format({'Total POIs': '{:.0f}', 'Mean': '{:.3f}', 'Std': '{:.3f}',
             'Q1': '{:.3f}', 'Median': '{:.3f}', 'Q3': '{:.3f}', 'Max': '{:.3f}'})
    .background_gradient(subset=['Mean', 'Std', 'Max'], cmap=palette_cmap)
```

```
    .set_properties(**{
        'font-family': 'Inter, sans-serif',
        'font-size': '12px',
        'color': '#1B2632',
        'border-bottom': '1px solid #C9C1B1',
        'padding': '7px 12px'
    })
    .set_table_styles([
        {'selector': 'thead tr',
         'props': [('background-color', '#2C3B4D'),
                   ('color', '#EEE9DF'),
                   ('font-weight', '500')]},
        {'selector': 'caption',
         'props': [('font-size', '11px'),
                   ('color', '#C9C1B1'),
                   ('font-style', 'italic'),
                   ('text-align', 'left'),
                   ('padding', '6px 0')]}
    ])
)
# Note: the high max values for some categories (e.g. cafe, restaurant) indicate that a few wards have very
# likely due to major commercial centers. This skew is important to note when interpreting the data and may
# investigation or transformation in future analyses.
```

Out[72]:

	Total POIs	Mean	Std	Q1	Median	Q3	Max
atm_density	1629	1.866	2.908	0.000	0.739	2.332	20.721
bank_density	643	0.722	1.866	0.000	0.000	0.333	12.158
bar_density	1098	1.372	3.971	0.000	0.000	1.052	47.284
cafe_density	6482	7.582	11.638	0.978	3.226	9.320	99.362
clothes_density	2861	3.547	13.987	0.000	0.000	1.760	191.327
convenience_density	5392	6.256	6.350	1.621	3.904	9.195	35.078
department_store_density	133	0.132	0.483	0.000	0.000	0.000	5.052
electronics_density	300	0.338	0.839	0.000	0.000	0.252	8.769
fast_food_density	5909	6.664	9.457	1.154	3.086	8.116	61.386
jewelry_density	678	0.782	3.937	0.000	0.000	0.074	79.490
mall_density	81	0.102	0.390	0.000	0.000	0.000	3.155
nightclub_density	150	0.195	0.788	0.000	0.000	0.000	9.632
pharmacy_density	1268	1.377	1.662	0.314	0.882	1.900	14.568
pub_density	3047	3.472	5.393	0.575	1.375	4.091	46.971
restaurant_density	8686	10.235	19.567	1.015	3.485	11.093	235.109
station_density	599	0.600	0.858	0.000	0.305	0.914	5.022
supermarket_density	1246	1.333	1.785	0.000	0.708	1.845	13.470

Table 5: POI Density Statistics by Category — Ward-Level Distribution Across Greater London. *Ward-level POI densities (POIs per km²). Restaurant and café densities show the highest means and greatest skew — restaurant_density peaks at 235.1, clothes_density at 191.3, both far exceeding their Q3 values, indicating concentration in a small number of central wards. Sparse categories (mall, department_store, nightclub) have a median of zero, absent from the majority of wards.*

Feature Engineering:

All crime and POI counts were normalised by ward area (per km²) to account for variation in ward size. We then applied a log transformation to all density variables to reduce right skewness before modelling, as shown in Tables 3 and 5.

$$x' = \log\left(\frac{\text{count}_i}{\text{area}_i} + 1\right)$$

where x' is the transformed feature, count_i is the raw crime or POI count for ward i, area_i is the ward area in km², and the $+1$ constant prevents undefined values where counts are zero. After transformation, residual skewness fell below 1.5 for 12 of the 17 POI features — mall, electronics, department_store, nightclub, and jewellery remaining moderately skewed due to their sparse distribution across wards. Figures 1 and 2 show the effect of this transformation on crime density and POI feature distributions respectively. Table 6 summarises all variables used in the model.

In [73]:

```
# Feature engineering
# log transform POI densities to reduce skew (add small constant to avoid log(0))
# ==============================================================================
# Figure 1: Effect of log transformation on crime density
```

```
# Figure 2: Skewness before and after log transformation
# Figure 3: Spatial distribution of crime and amenity densities
```

In [74]:
```
# Log transform all densities
all_density_cols = ["typeA_density", "typeB_density"] + poi_density_cols

for col in all_density_cols:
    ward_gdf[f"{col}_log"] = np.log1p(ward_gdf[col])

log_feature_cols = [f"{c}_log" for c in poi_density_cols]
log_targetA = "typeA_density_log"
log_targetB = "typeB_density_log"

# Before vs after skewness comparison
skew_before = ward_gdf[poi_density_cols].skew().rename("Before")
skew_after  = ward_gdf[log_feature_cols].skew().rename("After")
skew_after.index = skew_before.index  # align index for comparison

skew_df = pd.concat([skew_before, skew_after], axis=1).sort_values("Before", ascending=False)
```

In [75]:
```
# Distribution plots: raw vs log transformed
fig, axes = plt.subplots(2, 2, figsize=(12, 8), facecolor='#F7F4ED')

# We will compare the distributions of the raw crime density features (typeA_density, typeB_density) with th
pairs = [
    ("typeA_density", "typeA_density_log",
     "Type A — raw (crimes per km²)", "Type A — log transformed"),
    ("typeB_density", "typeB_density_log",
     "Type B — raw (crimes per km²)", "Type B — log transformed"),
]

# Create log(x+1) features for crime densities
for row, (raw, log, raw_title, log_title) in enumerate(pairs):
    for col, (col_name, title, color) in enumerate([
        (raw, raw_title, '#A35139'),
        (log, log_title, '#2C3B4D')
    ]):
        ax = axes[row][col]
        ax.set_facecolor('#F7F4ED')

        ward_gdf[col_name].hist(
            bins=40, ax=ax,
            color=color, edgecolor='#F7F4ED', linewidth=0.3
        )

        ax.set_title(title, fontsize=11, fontweight='bold',
                     color='#1B2632', pad=8)
        ax.set_xlabel('crimes per km²' if col == 0 else 'log(x+1)',
                      fontsize=9, color='#1B2632')
        ax.set_ylabel('Frequency', fontsize=9, color='#1B2632')
        ax.tick_params(colors='#1B2632', labelsize=8)
        ax.spines[['top', 'right']].set_visible(False)
        ax.spines[['left', 'bottom']].set_color('#C9C1B1')
        ax.yaxis.grid(True, color='#C9C1B1', linewidth=0.5, linestyle='--')
        ax.set_axisbelow(True)

fig.suptitle('Effect of log(x+1) transformation on crime density distributions',
             fontsize=13, fontweight='bold', color='#1B2632', y=1.02)


plt.tight_layout()
plt.show()
```

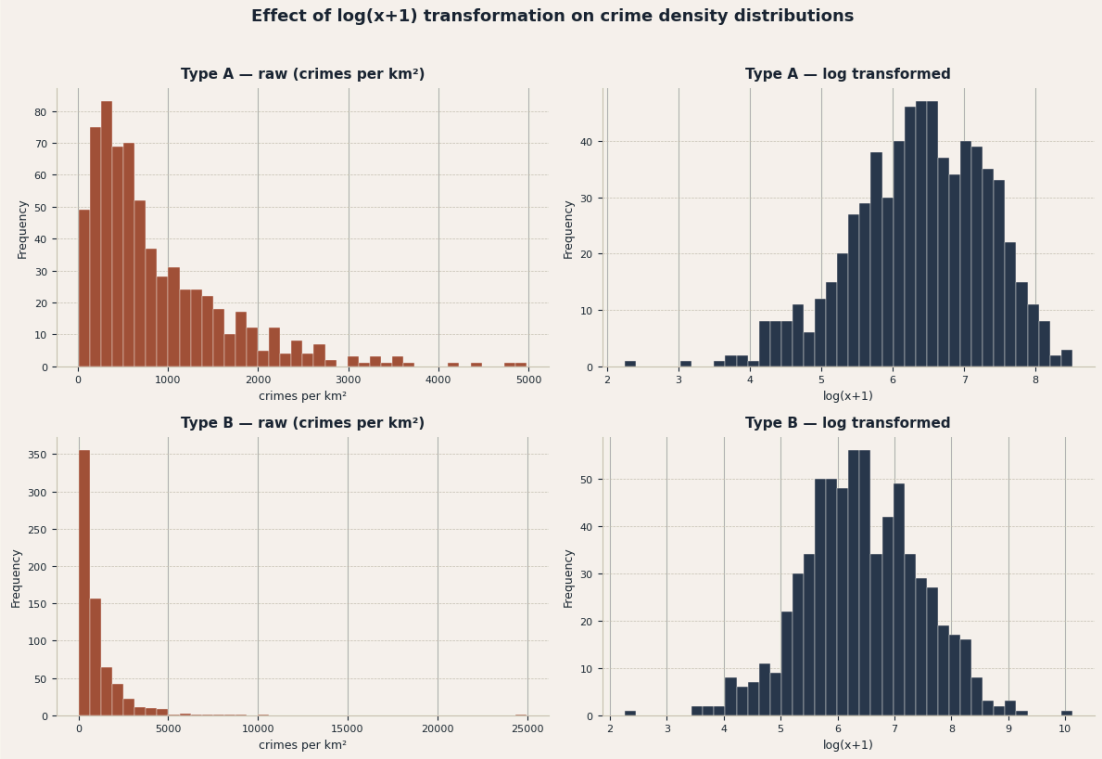


Figure 1: Effect of log(x+1) Transformation on Crime Density Distributions. *Raw (left) and log-transformed (right) distributions for Type A and Type B across 679 wards. Both raw distributions are strongly right-skewed — Type B notably more extreme (max ≈ 25,000 crimes per km²). log(x+1) produces approximately symmetric, unimodal distributions suitable for regression modelling.*

In [76]:
```
fig, axes = plt.subplots(1, 2, figsize=(12, 5), facecolor='#F7F4ED')

for ax, col, color, title in zip(
    axes,
    ["Before", "After"],
    ["#A35139", "#2C3B4D"],
    ["Before log transform", "After log transform"]
):
    ax.set_facecolor('#EEE9DF')
    bars = ax.barh(skew_df.index, skew_df[col],
                   color=color, edgecolor='none')
    ax.axvline(0, color='#1B2632', linewidth=0.8, linestyle='--')
    ax.axvline(1.5, color='#FFB162', linewidth=0.8,
               linestyle='--', label='Skewness = 1.5')
    ax.axvline(-1.5, color='#FFB162', linewidth=0.8, linestyle='--')
    ax.set_title(title, fontsize=12, fontweight='bold',
                 color='#1B2632', pad=10)
    ax.set_xlabel('Skewness', fontsize=10, color='#1B2632')
    ax.tick_params(colors='#1B2632', labelsize=9)
    ax.spines[['top', 'right']].set_visible(False)
    ax.spines[['left', 'bottom']].set_color('#C9C1B1')
    ax.xaxis.grid(True, color='#C9C1B1', linewidth=0.5, linestyle='--')
    ax.set_axisbelow(True)
    if col == "After":
        ax.set_xlim(-0.5, skew_df["After"].max() + 0.5)
    else:
        ax.set_xlim(0, skew_df["Before"].max() + 0.5)
    ax.legend(fontsize=8, framealpha=0)

fig.suptitle('Skewness before and after log(x+1) transformation',
             fontsize=13, fontweight='bold', color='#1B2632', y=1.02)

plt.tight_layout()
plt.show()
```

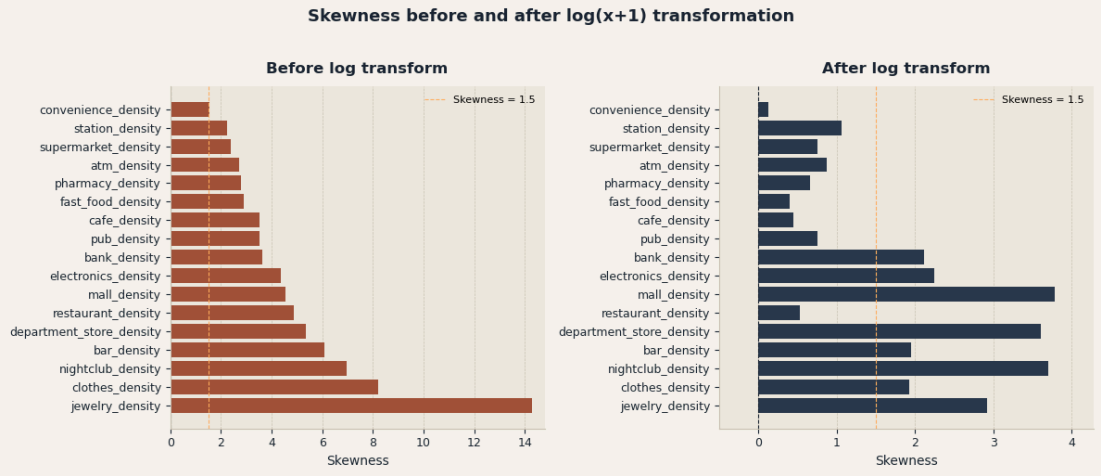


Figure 2: Skewness Before and After log(x+1) Transformation — POI Density Features *Pre-transformation, all 17 features exceed the ±1.5 threshold — jewelry_density most severely (skewness ≈ 14). Post-transformation, 12 fall within the acceptable range. The five remaining (mall, electronics, nightclub, department store, jewelry) retain elevated skew due to zero-inflation across outer-London wards, noted as a modelling limitation.*

In [77]:
```
# Save feature matrix
save_cols = (
    ["WardCode", "WardName", "Borough", "area_km2",
     "population",
     "typeA_count", "typeB_count",
     "typeA_density", "typeB_density"]
    + poi_density_cols
    + log_feature_cols
    + [log_targetA, log_targetB]
)

ward_gdf[save_cols].to_csv("data/feature_matrix.csv", index=False)
```

In [78]:
```
# Load feature matrix
feature_matrix = pd.read_csv(BASE + "feature_matrix.csv")
print(f"Feature matrix loaded — {feature_matrix.shape[0]} wards × {feature_matrix.shape[1]} columns")
feature_matrix.head(5)
```

```
Feature matrix loaded — 679 wards × 45 columns
```

Out[78]:

	WardCode	WardName	Borough	area_km2	population	typeA_count	typeB_count	typeA_density	typeB_density
0	E05009317	Bethnal Green	Tower Hamlets	1.209073	21282	2811	2403	2324.922210	1987.473522
1	E05009318	Blackwall & Cubitt Town	Tower Hamlets	1.352521	21328	1776	1615	1313.103530	1194.066555
2	E05009319	Bow East	Tower Hamlets	1.868250	19723	2085	1973	1116.017694	1056.068542
3	E05009320	Bow West	Tower Hamlets	1.358473	13954	1422	1457	1046.763438	1072.527658
4	E05009321	Bromley North	Tower Hamlets	0.600058	11753	1482	1403	2469.762368	2338.108368

5 rows × 45 columns

Table 6: Variable definitions, sources, units, and roles in the analysis

Variable	Type	Source	Unit	Role
typeA_density	Continuous	MPS / London Datastore	crimes per km²	Target — Type A
typeB_density	Continuous	MPS / London Datastore	crimes per km²	Target — Type B
restaurant_density	Continuous	OpenStreetMap	POIs per km²	Feature
cafe_density	Continuous	OpenStreetMap	POIs per km²	Feature
fast_food_density	Continuous	OpenStreetMap	POIs per km²	Feature
convenience_density	Continuous	OpenStreetMap	POIs per km²	Feature
clothes_density	Continuous	OpenStreetMap	POIs per km²	Feature
atm_density	Continuous	OpenStreetMap	POIs per km²	Feature
pub_density	Continuous	OpenStreetMap	POIs per km²	Feature

Variable	Type	Source	Unit	Role
pharmacy_density	Continuous	OpenStreetMap	POIs per km²	Feature
bar_density	Continuous	OpenStreetMap	POIs per km²	Feature
supermarket_density	Continuous	OpenStreetMap	POIs per km²	Feature
station_density	Continuous	OpenStreetMap	POIs per km²	Feature
bank_density	Continuous	OpenStreetMap	POIs per km²	Feature
jewelry_density	Continuous	OpenStreetMap	POIs per km²	Feature
electronics_density	Continuous	OpenStreetMap	POIs per km²	Feature
nightclub_density	Continuous	OpenStreetMap	POIs per km²	Feature
department_store_density	Continuous	OpenStreetMap	POIs per km²	Feature
mall_density	Continuous	OpenStreetMap	POIs per km²	Feature
area_km2	Continuous	ONS / EPSG:27700	km²	Normalisation denominator
population	Continuous	ONS mid-2021	persons	Context variable

*Note: All log-transformed versions of the above density variables (suffix _log) are used as model inputs.

In [79]:
```
# Choropleth maps — crime and POI densities
from matplotlib.colors import LinearSegmentedColormap

fig, axes = plt.subplots(1, 3, figsize=(16, 6), facecolor='#F7F4ED')

cmap_typeA = LinearSegmentedColormap.from_list('typeA', ['#F7F4ED', '#A35139'])
cmap_typeB = LinearSegmentedColormap.from_list('typeB', ['#F7F4ED', '#2C3B4D'])
cmap_poi   = LinearSegmentedColormap.from_list('poi',   ['#F7F4ED', '#FFB162'])

plots = [
    ("typeA_density",     "Person & Society Crime\n(crimes per km²)", cmap_typeA, "a"),
    ("typeB_density",     "Property Crime\n(crimes per km²)",          cmap_typeB, "b"),
    ("fast_food_density", "Fast Food\n(POIs per km²)",         cmap_poi,   "c"),
]

ward_plot = ward_gdf.to_crs("EPSG:27700")
gla_plot  = gla_gdf.to_crs("EPSG:27700")

minx, miny, maxx, maxy = gla_plot.total_bounds
pad_x = (maxx - minx) * 0.02
pad_y = (maxy - miny) * 0.02

for ax, (col, title, cmap, label) in zip(axes, plots):
    ax.set_facecolor('#F7F4ED')
    vmax = ward_plot[col].quantile(0.95)

    gla_plot.plot(ax=ax, color='#D8D3C8', edgecolor='none')

    ward_plot.plot(
        column=col,
        cmap=cmap,
        legend=True,
        vmin=0,
        vmax=vmax,
        linewidth=0.1,
        edgecolor='#C9C1B1',
        legend_kwds={
            "shrink": 0.8,
            "aspect": 25,
            "pad": 0.02,
            "format": "%.0f"
        },
        ax=ax,
        missing_kwds={"color": "#C9C1B1"}
    )

    gla_plot.boundary.plot(ax=ax, color='#1B2632', linewidth=1.0)

    # Style the colorbar geopandas created
    cbar = ax.get_figure().axes[-1]
    cbar.tick_params(labelsize=8, colors='#1B2632')
    cbar.set_facecolor('#F7F4ED')
    for spine in cbar.spines.values():
        spine.set_visible(False)

    ax.set_xlim(minx - pad_x, maxx + pad_x)
    ax.set_ylim(miny - pad_y, maxy + pad_y)

    ax.set_title(title, fontsize=10, fontweight='bold',
                 color='#1B2632', pad=8)
    ax.text(0.02, 0.98, label, transform=ax.transAxes,
            fontsize=11, fontweight='bold', color='#1B2632',
            va='top', ha='left')
```

```
    ax.set_axis_off()

fig.suptitle('Crime and amenity density across London wards',
             fontsize=13, fontweight='bold', color='#1B2632', y=1.01)

plt.tight_layout()
plt.show()
```

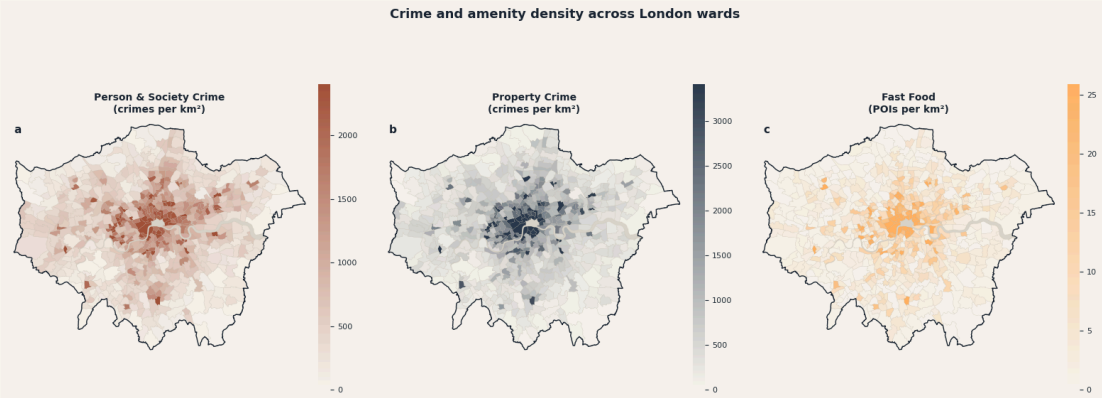


Figure 3: Spatial distribution of crime and amenity density across London wards *(a) Type A Crime - person and society density; (b) Type B Crime - property density; (c) fast food POI density as a representative amenity feature. All values per km², capped at the 95th percentile. Both crime types show strong centre-periphery gradients, with the highest densities in inner-London boroughs — fast-food density closely mirrors this spatial pattern.*

With the feature matrix prepared — 679 wards × 17 log-transformed POI density features — four models of increasing complexity were applied to predict crime density across London wards.

Methodology

[go back to the top]

Analytical Framework:

The methodology is structured as a progression of analytical layers. We begin with Multiple Linear Regression (MLR), which assumes the simplest possible relationship between POI density and crime; a straight line that assumes each amenity type contributes independently and additively. K-Nearest Neighbors then asks whether wards with similar amenity profiles experience similar crime levels. Lastly, Random Forest removes all these assumptions, building hundreds of decision trees across random subsets of the data to capture the complex, non-linear interactions between amenity types that the previous two methods cannot. Having established what the model predicts, SHAP values further evaluate it by decomposing each prediction into the individual contributions of each amenity type, showing not just which features matter, but also how much and in which direction.

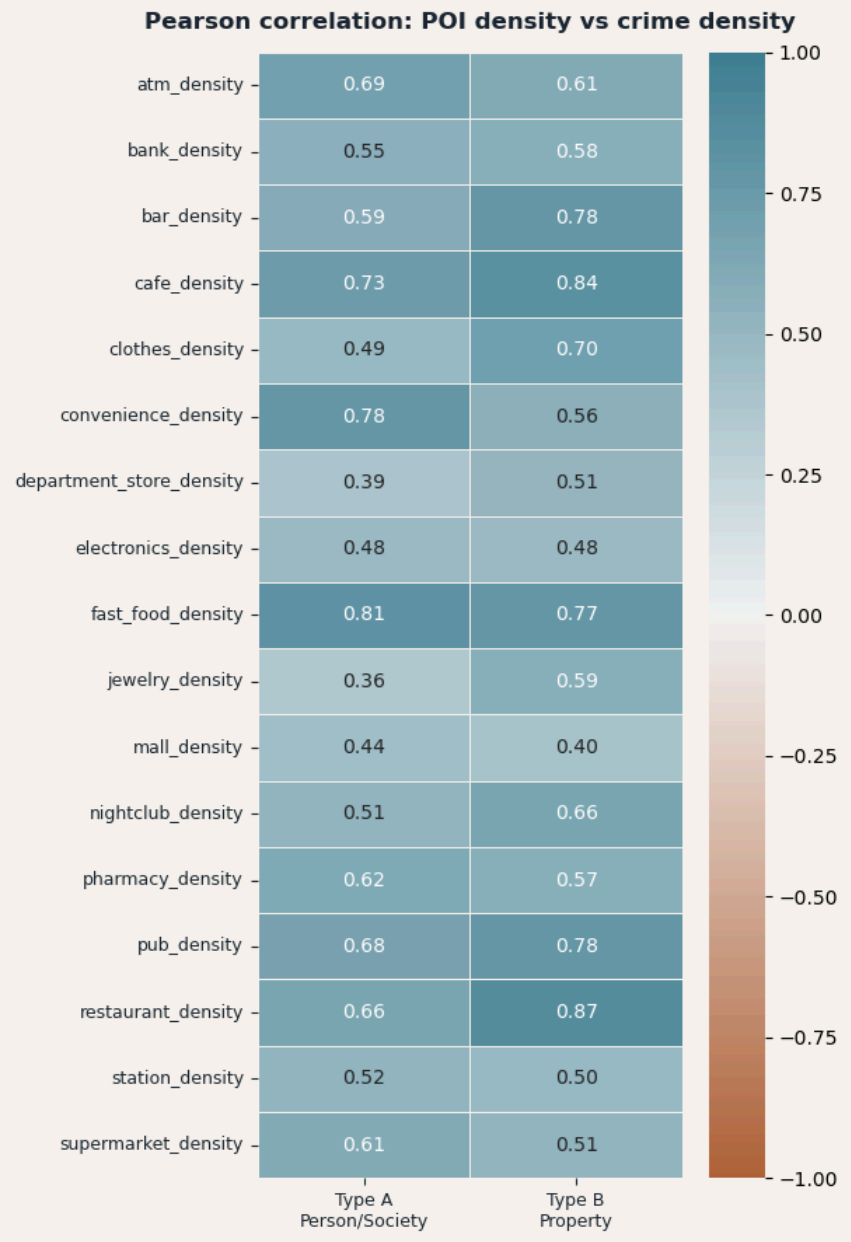
Pre-Modelling Diagnostics:

Before fitting any model, three diagnostic checks were conducted to assess feature behaviour and justify model selection. Figures 4, 5, and 6 show the results.

In [80]:
```
# Pre-modelling diagnostics
# ==========================================================================================
# Figure 4:Pearson Correlation Between POI Density and Crime Density by Category
# Figure 5: Pearson Intercorrelation Matrix for POI Density Features
# Figure 6: Linearity Check – Top POI Features vs Type A Crime Density
```

In [81]:
```
# Correlation heatmap: POI density vs crime targets

# We will compute the Pearson correlation coefficients between each POI density feature and the two crime de
poi_density_cols = [c for c in feature_matrix.columns
                    if c.endswith('_density')
                    and c not in ['typeA_density', 'typeB_density']]

# We will focus on the raw density features for correlation analysis, as the log transformation is primarily
target_cols  = ["typeA_density", "typeB_density"]

# Compute correlation matrix and extract relevant correlations
corr = feature_matrix[poi_density_cols + target_cols].corr()
corr_targets = corr.loc[poi_density_cols, target_cols]

# Heatmap of correlations
fig, ax = plt.subplots(figsize=(6, 9), facecolor='#F7F4ED')
ax.set_facecolor('#F7F4ED')

sns.heatmap(
```

```python
    corr_targets,
    annot=True, fmt=".2f",
    cmap=sns.diverging_palette(30, 220, as_cmap=True),
    center=0,
    vmin=-1, vmax=1,
    xticklabels=["Type A\nPerson/Society", "Type B\nProperty"],
    linewidths=0.5,
    linecolor='#F7F4ED',
    ax=ax
)

ax.set_title("Pearson correlation: POI density vs crime density",
             fontsize=12, fontweight='bold', color='#1B2632', pad=12)
ax.tick_params(colors='#1B2632', labelsize=9)

plt.tight_layout()
plt.show()

print("\nTop predictors — Type A (person/society crime):")
print(corr_targets["typeA_density"].sort_values(ascending=False).round(2).head(5))
print("\nTop predictors — Type B (property crime):")
print(corr_targets["typeB_density"].sort_values(ascending=False).round(2).head(5))
```

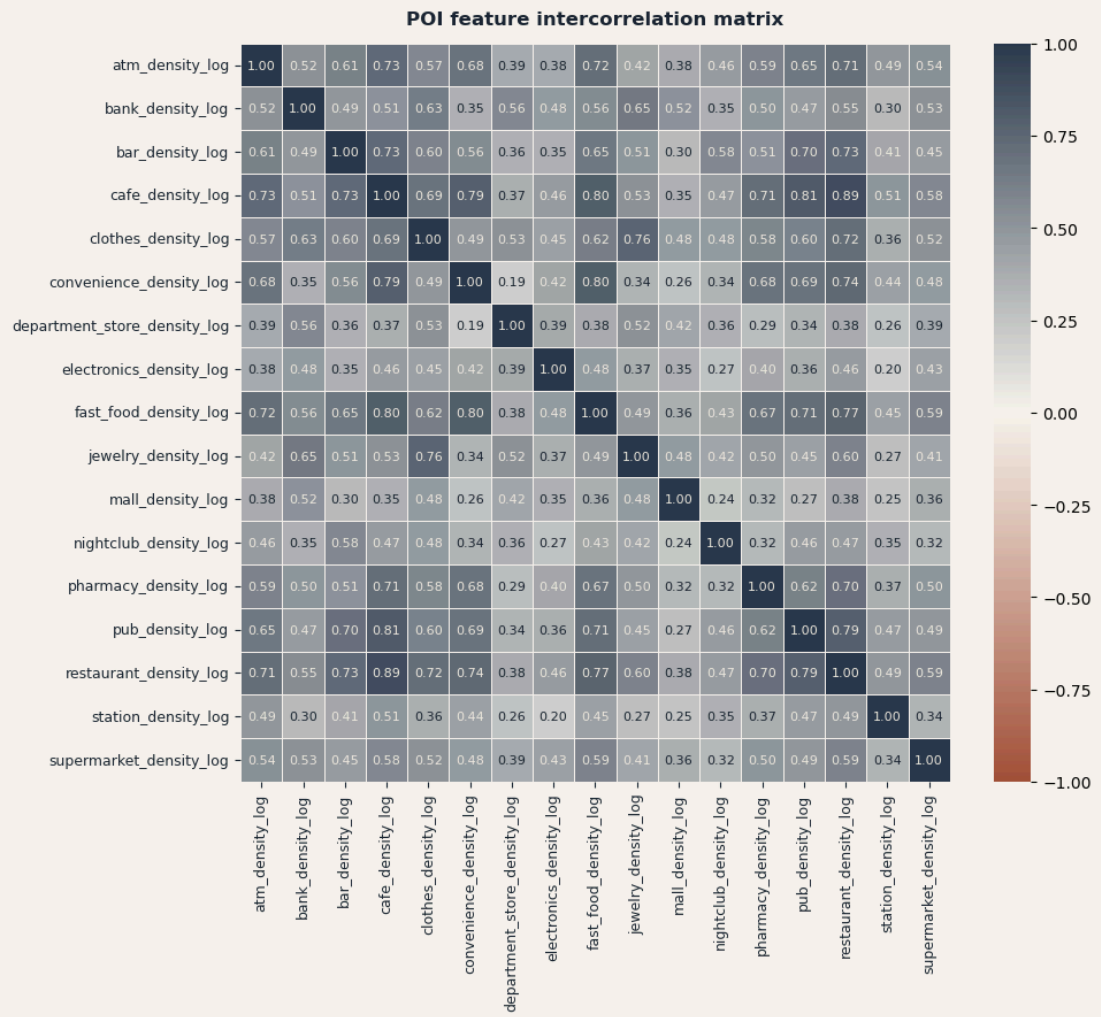



```
Top predictors — Type A (person/society crime):
fast_food_density      0.81
convenience_density    0.78
cafe_density           0.73
atm_density            0.69
pub_density            0.68
Name: typeA_density, dtype: float64

Top predictors — Type B (property crime):
restaurant_density    0.87
cafe_density          0.84
pub_density           0.78
bar_density           0.78
fast_food_density     0.77
Name: typeB_density, dtype: float64
```

Figure 4: Pearson Correlation Between POI Density and Crime Density by Category *Pearson correlation coefficients between log-transformed POI density features and crime density targets across 679 London*

wards. Fast food (r=0.81) and convenience store (r=0.78) are the strongest correlates of Type A crime; restaurant (r=0.87) and café (r=0.84) lead for Type B.

In [82]:
```
# Multicollinearity check between POI features
poi_corr = feature_matrix[log_feature_cols].corr()

# Heatmap of POI feature intercorrelations with adaptive annotations
fig, ax = plt.subplots(figsize=(10, 9), facecolor='#F7F4ED')
ax.set_facecolor('#F7F4ED')

# We will compute the Pearson correlation coefficients between the log-transformed POI density features to c
from matplotlib.colors import LinearSegmentedColormap
div_cmap = LinearSegmentedColormap.from_list(
    'div', ['#A35139', '#F7F4ED', '#2C3B4D']
)

# Create heatmap without annotations to allow for custom annotation with adaptive text color
im = sns.heatmap(
    poi_corr,
    annot=False,          # turn off seaborn annotations
    cmap=div_cmap, center=0,
    vmin=-1, vmax=1,
    linewidths=0.5,
    linecolor='#F7F4ED',
    ax=ax
)

# Manual annotations with adaptive colour
for i in range(len(poi_corr)):
    for j in range(len(poi_corr.columns)):
        val = poi_corr.iloc[i, j]
        # White text for dark cells, dark text for light/mid cells
        text_color = '#EEE9DF' if abs(val) > 0.4 else '#1B2632'
        ax.text(
            j + 0.5, i + 0.5, f"{val:.2f}",
            ha='center', va='center',
            fontsize=8, color=text_color
        )

ax.set_title('POI feature intercorrelation matrix',
             fontsize=12, fontweight='bold', color='#1B2632', pad=12)
ax.tick_params(colors='#1B2632', labelsize=9)


plt.tight_layout()
plt.show()

# Flag highly correlated pairs
high_corr = [
    (poi_corr.columns[i], poi_corr.columns[j], round(poi_corr.iloc[i, j], 2))
    for i in range(len(poi_corr.columns))
    for j in range(i + 1, len(poi_corr.columns))
    if abs(poi_corr.iloc[i, j]) > 0.8
]

print("Highly correlated POI pairs (r > 0.8):")
for a, b, r in high_corr:
    print(f"  {a} × {b}: r = {r}")
```




```
Highly correlated POI pairs (r > 0.8):
  cafe_density_log × fast_food_density_log: r = 0.8
  cafe_density_log × pub_density_log: r = 0.81
  cafe_density_log × restaurant_density_log: r = 0.89
```

Figure 5: POI feature intercorrelation matrix *Pairwise Pearson correlation coefficients between all 17 log-transformed POI density features. Three pairs exceed r=0.80: café × restaurant (r=0.89), café × pub (r=0.81), and café × fast food (r=0.80), reflecting the spatial co-occurrence of food and beverage amenities in inner-London wards. This multicollinearity violates MLR independence assumptions and justifies Random Forest as the primary model.*

In [83]:

```
# Linearity check: top POI features vs crime targets
fig, axes = plt.subplots(2, 3, figsize=(14, 8), facecolor='#F7F4ED')

# We will create scatter plots of the top POI density features (based on correlation with type A crime densi
top_features = [
    "convenience_density_log", "fast_food_density_log",
    "restaurant_density_log", "cafe_density_log",
    "atm_density_log", "pub_density_log"
]

# We will plot the log-transformed POI density features against the log-transformed type A crime density to
for ax, feat in zip(axes.flatten(), top_features):
    ax.set_facecolor('#F7F4ED')

    ax.scatter(
        feature_matrix[feat],
        feature_matrix["typeA_density_log"],
        alpha=0.25, s=12, color='#A35139', edgecolors='none'
    )

    # Add lowess trend line
    from statsmodels.nonparametric.smoothers_lowess import lowess
    sorted_x = feature_matrix[feat].sort_values()
    trend = lowess(
        feature_matrix["typeA_density_log"],
        feature_matrix[feat],
        frac=0.4
    )
    ax.plot(trend[:, 0], trend[:, 1],
            color='#2C3B4D', linewidth=1.5)

    label = feat.replace("_density_log", "").replace("_", " ")
    ax.set_xlabel(f'{label} density (log)', fontsize=9, color='#1B2632')
    ax.set_ylabel('Type A crime (log)', fontsize=9, color='#1B2632')
    ax.set_title(label.title(), fontsize=10, fontweight='bold',
                 color='#1B2632', pad=6)
    ax.tick_params(colors='#1B2632', labelsize=8)
```

```
    ax.spines[['top', 'right']].set_visible(False)
    ax.spines[['left', 'bottom']].set_color('#C9C1B1')

fig.suptitle('Linearity check: top POI features vs Type A crime density',
             fontsize=13, fontweight='bold', color='#1B2632', y=1.02)


plt.tight_layout()
plt.show()
```

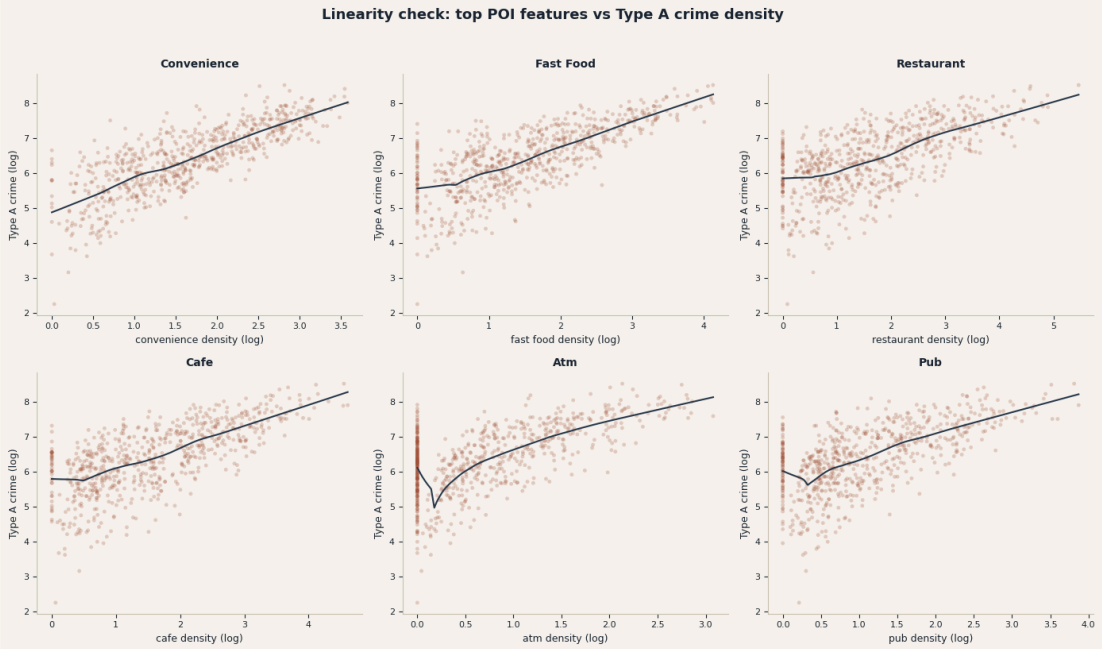


Figure 6: Linearity Check — Top POI Features vs Type A Crime Density *Scatter plots with LOWESS trend lines across 679 London wards. Convenience, fast food, restaurants, and cafés display broadly linear trends, while ATMs and pubs exhibit notable nonlinearity — flat at low densities before rising sharply. This confirms that a linear model would misrepresent several key relationships among features.*

Together, these diagnostics confirm strong POI-crime associations, significant multicollinearity among food and beverage features, and non-linear relationships for several predictors — collectively motivating the use of Random Forest as the primary model.

Methodology:

The dataset was split 80/20 into training (543 wards) and test (136 wards) sets using a fixed random seed (42) for reproducibility. To confirm that test-set performance is not sensitive to the particular train-test partition, 5-fold cross-validation was applied to the full dataset using the same hyperparameters.

1. **MLR** — Features were standardised to enable direct coefficient comparison. Included as an interpretable baseline despite known limitations — multicollinearity and non-linearity confirmed in Figures 5 and 6. Sign reversals on café and restaurant coefficients are multicollinearity artefacts, not errors (Figure 7).

2. **KNN** — Tests whether wards with similar amenity profiles experience similar crime levels, making no assumptions about functional form. k=4 was selected via 5-fold cross-validation averaged across both targets (Figure 9).

3. **RF** — Removes linearity and independence assumptions entirely, capturing non-linear interactions between amenity types. Selected as the primary model on the basis of pre-modelling diagnostics. Hyperparameters tuned via grid search: n_estimators=300, max_depth=10, min_samples_leaf=4 (CV R^2=0.683).

4. **SHAP** — Decomposes each ward-level RF prediction into individual feature contributions, enabling direct comparison of which amenity types drive each crime category and in which direction.

Figures 7 and 8 show the standardised MLR coefficients for Type A and Type B, respectively, illustrating the multicollinearity artefacts discussed above.

In [84]:
```
# Mythodology section
# Split data into training and test sets
# ==============================================================================
# Figure 7+8: MLR Standardised Coefficients — Type A Person and Society Crime + Type B Property Crime
# Figure 9: KNN — Cross-Validated R² by k (Mean of Both Targets)
# Table 7: Model Performance on Held-Out Test Set
```

```
# Figure 10: Model Performance Comparison — R² on Held-Out Test Set
# Figure 11: SHAP Feature Contributions — Type A Person and Society Crime
# Figure 12: SHAP Feature Contributions — Type B Property Crime
```

In [85]:

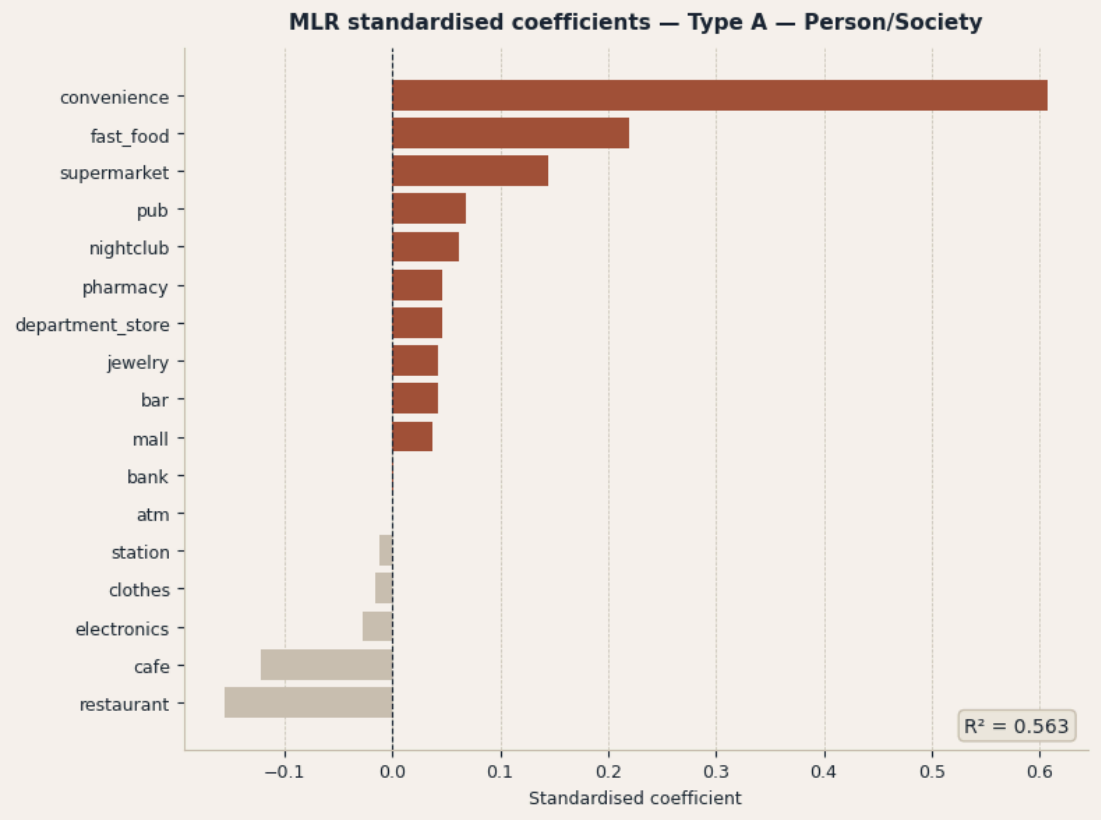
```
# Feature setup + train/test split
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import r2_score, mean_squared_error
import warnings
warnings.filterwarnings("ignore")

# Features and targets — drawn from feature matrix
log_feature_cols = [c for c in feature_matrix.columns
                    if c.endswith('_density_log')
                    and c not in ['typeA_density_log', 'typeB_density_log']]

feature_cols = log_feature_cols
targetA      = "typeA_density_log"
targetB      = "typeB_density_log"

# Drop any rows with missing values
model_df = feature_matrix[feature_cols + [targetA, targetB]].dropna().copy()

X  = model_df[feature_cols]
yA = model_df[targetA]
yB = model_df[targetB]

X_train, X_test, yA_train, yA_test, yB_train, yB_test = train_test_split(
    X, yA, yB, test_size=0.2, random_state=42
)

# Standardise for MLR and KNN (RF does not require this)
scaler = StandardScaler()
X_train_sc = scaler.fit_transform(X_train)
X_test_sc  = scaler.transform(X_test)

print(f"Modelling dataset: {len(model_df)} wards")
print(f"Train: {X_train.shape[0]} wards | Test: {X_test.shape[0]} wards")
print(f"Features: {len(feature_cols)}")
print(f"\nFeature list: {[f.replace('_density_log', '') for f in feature_cols]}")
```

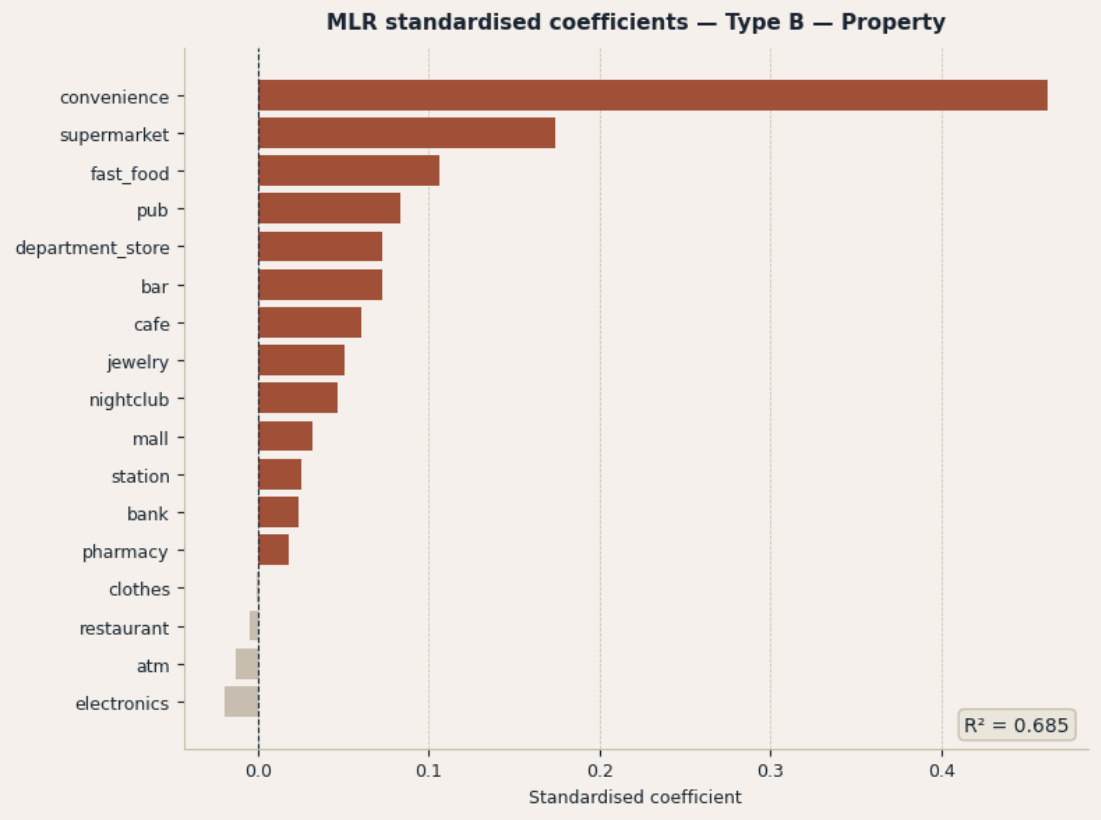
```
Modelling dataset: 679 wards
Train: 543 wards | Test: 136 wards
Features: 17

Feature list: ['atm', 'bank', 'bar', 'cafe', 'clothes', 'convenience', 'department_store', 'electronics', 'f
ast_food', 'jewelry', 'mall', 'nightclub', 'pharmacy', 'pub', 'restaurant', 'station', 'supermarket']
```

In [86]:

```
# Multiple Linear Regression
from sklearn.linear_model import LinearRegression
from sklearn.metrics import root_mean_squared_error

results = {}

# We will fit a multiple linear regression model to predict the log-transformed crime density targets (typeA
# using the log-transformed POI density features. We will evaluate model performance using R²
# and RMSE on the test set, and we will also visualize the standardised coefficients to understand
# which features are most influential in predicting crime densities.
for target_name, y_train, y_test in [
    ("Type A — Person/Society", yA_train, yA_test),
    ("Type B — Property",       yB_train, yB_test)
]:
    mlr = LinearRegression()
    mlr.fit(X_train_sc, y_train)
    y_pred = mlr.predict(X_test_sc)

    r2   = r2_score(y_test, y_pred)
    rmse = root_mean_squared_error(y_test, y_pred)
    results[f"MLR — {target_name}"] = {"R²": round(r2, 3), "RMSE": round(rmse, 3)}

    # Coefficient plot
    coef_df = pd.DataFrame({
        "Feature":     [c.replace("_density_log", "") for c in feature_cols],
        "Coefficient": mlr.coef_
    }).sort_values("Coefficient", ascending=True)

    fig, ax = plt.subplots(figsize=(8, 6), facecolor='#F7F4ED')
    ax.set_facecolor('#F7F4ED')

    colors = ['#A35139' if c > 0 else '#C9C1B1' for c in coef_df["Coefficient"]]
    ax.barh(coef_df["Feature"], coef_df["Coefficient"],
            color=colors, edgecolor='none')
    ax.axvline(0, color='#1B2632', linewidth=0.8, linestyle='--')

    ax.set_title(f'MLR standardised coefficients — {target_name}',
                 fontsize=11, fontweight='bold', color='#1B2632', pad=10)
    ax.set_xlabel('Standardised coefficient', fontsize=9, color='#1B2632')
    ax.tick_params(colors='#1B2632', labelsize=9)
    ax.spines[['top', 'right']].set_visible(False)
    ax.spines[['left', 'bottom']].set_color('#C9C1B1')
    ax.xaxis.grid(True, color='#C9C1B1', linewidth=0.5, linestyle='--')
    ax.set_axisbelow(True)

    # R² annotation on plot
```

```python
    ax.text(0.98, 0.02, f'R² = {r2:.3f}',
            transform=ax.transAxes, ha='right', va='bottom',
            fontsize=10, color='#1B2632',
            bbox=dict(boxstyle='round,pad=0.3', facecolor='#EEE9DF',
                      edgecolor='#C9C1B1'))

    plt.tight_layout()
    plt.show()

# Summary
print("\nMLR Summary:")
for k, v in results.items():
    print(f"  {k}: R² = {v['R²']:.3f} | RMSE = {v['RMSE']:.3f}")
```



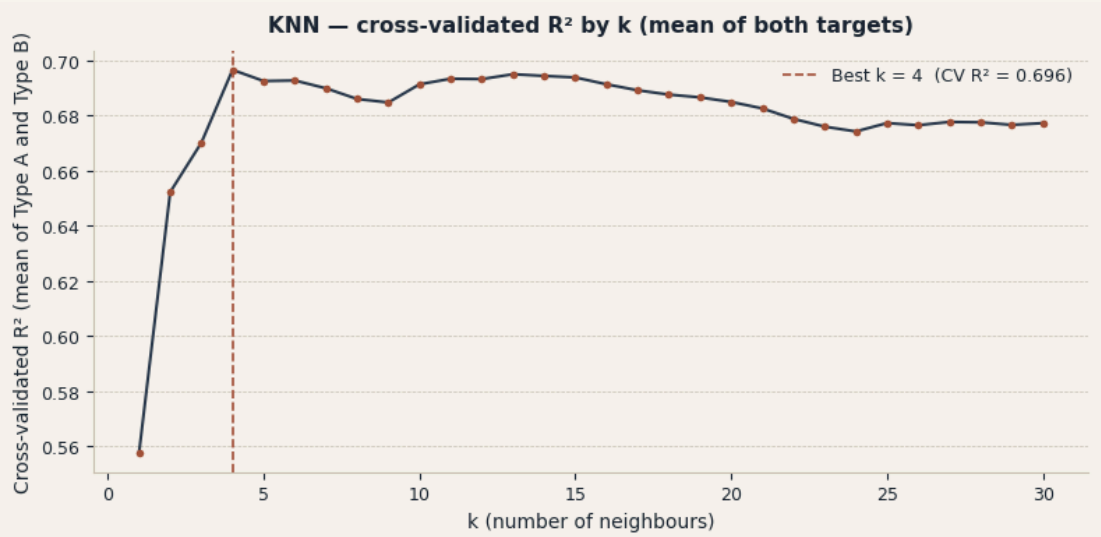



```
MLR Summary:
  MLR — Type A — Person/Society: R² = 0.563 | RMSE = 0.565
  MLR — Type B — Property: R² = 0.685 | RMSE = 0.540
```

Figure 7: MLR Standardised Coefficients — Type A Person and Society Crime *Standardised coefficients for Type A crime (R^2=0.563). Convenience store is the dominant predictor. Negative coefficients for restaurant and café are multicollinearity artefacts — both show positive SHAP values in the Random Forest model.*

Figure 8: MLR Standardised Coefficients — Type B Property Crime *Standardised coefficients for Type B crime (R^2=0.685). Convenience store again dominates, with supermarket and fast food as secondary predictors. Negative coefficients for restaurant and ATM reflect multicollinearity rather than genuine inverse relationships.*

In [87]:
```
#  KNN Regressor (tune k)
from sklearn.neighbors import KNeighborsRegressor
from sklearn.model_selection import cross_val_score

# Tune k using average of Type A and Type B CV R²
k_scores = []
for k in range(1, 31):
    knn = KNeighborsRegressor(n_neighbors=k)
    scoreA = cross_val_score(knn, X_train_sc, yA_train, cv=5, scoring="r2").mean()
    scoreB = cross_val_score(knn, X_train_sc, yB_train, cv=5, scoring="r2").mean()
    k_scores.append((scoreA + scoreB) / 2)

best_k = k_scores.index(max(k_scores)) + 1

# K tuning plot
fig, ax = plt.subplots(figsize=(8, 4), facecolor='#F7F4ED')
ax.set_facecolor('#F7F4ED')

ax.plot(range(1, 31), k_scores,
        color='#2C3B4D', linewidth=1.5, marker='o',
        markersize=4, markerfacecolor='#A35139', markeredgewidth=0)
ax.axvline(best_k, color='#A35139', linestyle='--', linewidth=1.2,
           label=f'Best k = {best_k}  (CV R² = {max(k_scores):.3f})')

ax.set_xlabel('k (number of neighbours)', fontsize=10, color='#1B2632')
ax.set_ylabel('Cross-validated R² (mean of Type A and Type B)', fontsize=10, color='#1B2632')
ax.set_title('KNN — cross-validated R² by k (mean of both targets)',
             fontsize=11, fontweight='bold', color='#1B2632', pad=10)
ax.tick_params(colors='#1B2632', labelsize=9)
ax.spines[['top', 'right']].set_visible(False)
ax.spines[['left', 'bottom']].set_color('#C9C1B1')
ax.yaxis.grid(True, color='#C9C1B1', linewidth=0.5, linestyle='--')
ax.set_axisbelow(True)
ax.legend(fontsize=9, framealpha=0, labelcolor='#1B2632')

plt.tight_layout()
plt.show()

# Fit KNN with best k — both targets
for target_name, y_train, y_test in [
    ("Type A — Person/Society", yA_train, yA_test),
    ("Type B — Property",       yB_train, yB_test)
]:
    knn = KNeighborsRegressor(n_neighbors=best_k)
    knn.fit(X_train_sc, y_train)
    y_pred = knn.predict(X_test_sc)

    r2   = r2_score(y_test, y_pred)
    rmse = root_mean_squared_error(y_test, y_pred)

    results[f"KNN — {target_name}"] = {"R²": round(r2, 3), "RMSE": round(rmse, 3)}

print(f"\nKNN Summary (k={best_k}):")
for k, v in {k: v for k, v in results.items() if 'KNN' in k}.items():
    print(f"  {k}: R² = {v['R²']:.3f} | RMSE = {v['RMSE']:.3f}")
```




```
KNN Summary (k=4):
  KNN — Type A — Person/Society: R² = 0.458 | RMSE = 0.629
  KNN — Type B — Property: R² = 0.651 | RMSE = 0.568
```

Figure 9: KNN — Cross-Validated R^2 by k (Mean of Both Targets) *Cross-validated R^2 (mean of Type A and Type B) across k=1 to k=30. Performance peaks at k=4 (CV R^2=0.696) before plateauing and gradually*

declining. k=4 was selected for both models.

The following section presents model performance, SHAP-based feature interpretation, and spatial diagnostics.

Results and Discussion

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Model performance

As shown in Table 7, Random Forest outperformed both baseline methods across all targets and metrics. The consistent gap between RF and MLR indicates that the POI-crime relationship is non-linear, with amenity types interacting in ways that additive linear models cannot fully capture. The underperformance of KNN suggests that wards with similar POI profiles do not always exhibit similar crime levels, pointing to spatial heterogeneity in crime determinants across London's diverse ward typologies.

Property crime (Type B) was predicted more accurately than Type A across all three methods, consistent with property crime being more directly opportunity-driven and therefore more strongly determined by the physical configuration of targets.

In [88]:
```
#  Random Forest (tuned) + model comparison
from sklearn.ensemble import RandomForestRegressor

# We will fit a Random Forest regressor to predict the log-transformed crime density targets using the log-t
best_params = {
    "max_depth":        10,
    "min_samples_leaf": 4,
    "n_estimators":     300,
    "random_state":     42,
    "n_jobs":           -1
}

# Fit RF with best hyperparameters — both targets
rf_models = {}

# We will train separate Random Forest models for each crime density target (Type A and Type B) using the sa
for target_name, y_train, y_test in [
    ("Type A — Person/Society", yA_train, yA_test),
    ("Type B — Property",       yB_train, yB_test)
]:
    rf = RandomForestRegressor(**best_params)
    rf.fit(X_train, y_train)
    y_pred = rf.predict(X_test)

    r2   = r2_score(y_test, y_pred)
    rmse = root_mean_squared_error(y_test, y_pred)

    results[f"RF — {target_name}"] = {"R²": round(r2, 3), "RMSE": round(rmse, 3)}
    rf_models[target_name] = rf

# Model comparison table
comparison_rows = []
for key, vals in results.items():
    parts = key.split(" — ", 1)
    comparison_rows.append({
        "Method":  parts[0],
        "Target":  parts[1],
        "R²":      vals["R²"],
        "RMSE":    vals["RMSE"]
    })

comparison_df = pd.DataFrame(comparison_rows).sort_values(
    ["Target", "R²"], ascending=[True, False]
).reset_index(drop=True)

# Styled comparison table
(comparison_df.style
    .format({'R²': '{:.3f}', 'RMSE': '{:.3f}'})
    .highlight_max(subset=['R²'], color='#C9E8D1')
    .highlight_min(subset=['RMSE'], color='#C9E8D1')
    .set_properties(**{
        'font-family': 'Inter, sans-serif',
        'font-size': '13px',
        'color': '#1B2632',
        'border-bottom': '1px solid #C9C1B1',
        'padding': '8px 14px'
    })
    .set_table_styles([
        {'selector': 'thead tr',
         'props': [('background-color', '#2C3B4D'),
                   ('color', '#EEE9DF'),
                   ('font-weight', '500')]},
        {'selector': 'caption',
         'props': [('font-size', '11px'),
```

```
                    ('color', '#C9C1B1'),
                    ('font-style', 'italic'),
                    ('text-align', 'left'),
                    ('padding', '6px 0')]}
    ])
    .hide(axis='index')
)
```

Out[88]:

Method	Target	R^2	RMSE
RF	Type A — Person/Society	0.623	0.524
MLR	Type A — Person/Society	0.563	0.565
KNN	Type A — Person/Society	0.458	0.629
RF	Type B — Property	0.716	0.512
MLR	Type B — Property	0.685	0.540
KNN	Type B — Property	0.651	0.568

Table 7: Model Performance on Held-Out Test Set *R^2 and RMSE for all three models across both crime density targets (136 test wards). Random Forest outperforms both baselines. Property crime (Type B) is predicted more accurately than Type A across all methods.*

In [89]:

```
# Model comparison bar chart
fig, axes = plt.subplots(1, 2, figsize=(12, 5), facecolor='#F7F4ED')

method_colors = {
    'MLR': '#C9C1B1',
    'KNN': '#2C3B4D',
    'RF':  '#A35139'
}

for ax, target in zip(axes, [
    "Type A — Person/Society", "Type B — Property"
]):
    subset = comparison_df[comparison_df["Target"] == target].copy()
    colors = [method_colors.get(m, '#C9C1B1') for m in subset["Method"]]

    ax.set_facecolor('#F7F4ED')
    bars = ax.bar(subset["Method"], subset["R²"],
                  color=colors, width=0.5, edgecolor='none')

    # Value labels
    for bar in bars:
        height = bar.get_height()
        ax.text(bar.get_x() + bar.get_width() / 2,
                height + 0.01, f'{height:.3f}',
                ha='center', va='bottom',
                fontsize=10, color='#1B2632', fontweight='500')

    ax.set_ylim(0, 1)
    ax.axhline(0.5, color='#FFB162', linewidth=0.8,
               linestyle='--', label='R² = 0.5')
    ax.set_title(target, fontsize=11, fontweight='bold',
                 color='#1B2632', pad=10)
    ax.set_ylabel('R²', fontsize=10, color='#1B2632')
    ax.tick_params(colors='#1B2632', labelsize=10)
    ax.spines[['top', 'right']].set_visible(False)
    ax.spines[['left', 'bottom']].set_color('#C9C1B1')
    ax.yaxis.grid(True, color='#C9C1B1', linewidth=0.5, linestyle='--')
    ax.set_axisbelow(True)
    ax.legend(fontsize=9, framealpha=0, labelcolor='#1B2632')

fig.suptitle('Model performance comparison — R² on held-out test set',
             fontsize=13, fontweight='bold', color='#1B2632', y=1.02)

plt.tight_layout()
plt.show()
```

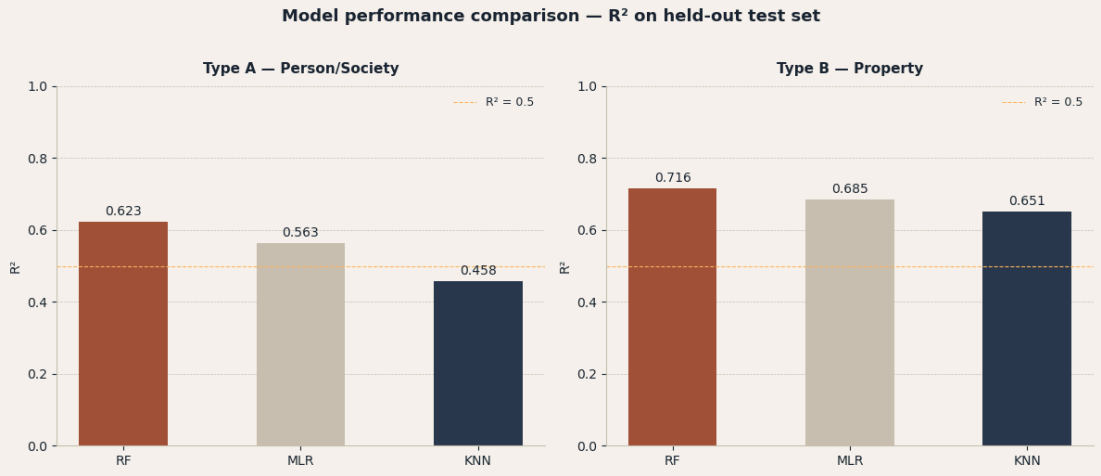


Figure 10: Model Performance Comparison — R² on Held-Out Test Set *R^2 scores for MLR, KNN, and Random Forest across both targets (136 test wards). The orange dashed line marks R^2=0.5. Random Forest achieves the highest R^2 for both Type A (0.623) and Type B (0.716). The spread between methods is larger for Type A — where KNN falls below 0.5 — than for Type B, where all three models perform comparably above 0.65.*

In [90]:

```
# RF CV validation
# 5-fold CV on the full dataset to validate that the 80/20 test-set R²
# is not an artefact of a fortunate split.

from sklearn.model_selection import cross_val_score
import numpy as np

cv_results = {}

for target_name, y_col in [
    ("Type A — Person/Society", "typeA_density_log"),
    ("Type B — Property",       "typeB_density_log"),
]:
    y = feature_matrix[y_col].values
    X = feature_matrix[feature_cols].fillna(0).values

    rf_cv = RandomForestRegressor(**best_params)
    scores = cross_val_score(rf_cv, X, y, cv=5, scoring="r2", n_jobs=-1)

    cv_results[target_name] = {
        "CV R² (mean)":  round(scores.mean(), 3),
        "CV R² (std)":   round(scores.std(),  3),
        "Test R²":       results[f"RF — {target_name}"]["R²"],
    }
    print(f"{target_name}")
    print(f"  CV R²:   {scores.mean():.3f} ± {scores.std():.3f}  (folds: {np.round(scores, 3)})")
    print(f"  Test R²: {cv_results[target_name]['Test R²']}\n")

# Styled summary table
cv_df = pd.DataFrame(cv_results).T.reset_index().rename(columns={"index": "Target"})

(cv_df.style
    .format({"CV R² (mean)": "{:.3f}", "CV R² (std)": "{:.3f}", "Test R²": "{:.3f}"})
    .set_properties(**{
        "font-family": "Inter, sans-serif",
        "font-size":   "13px",
        "color":       "#1B2632",
        "border-bottom": "1px solid #C9C1B1",
        "padding":     "8px 14px",
    })
    .set_table_styles([
        {"selector": "thead tr",
         "props": [("background-color", "#2C3B4D"),
                   ("color", "#EEE9DF"),
                   ("font-weight", "500")]},
        {"selector": "caption",
         "props": [("font-size", "11px"),
                   ("color", "#C9C1B1"),
                   ("font-style", "italic"),
                   ("text-align", "left"),
                   ("padding", "6px 0")]},
    ])
    .hide(axis="index")
)
```

```
Type A — Person/Society
  CV R²:   0.677 ± 0.064  (folds: [0.743 0.557 0.681 0.688 0.714])
  Test R²: 0.623

Type B — Property
  CV R²:   0.753 ± 0.084  (folds: [0.812 0.591 0.781 0.764 0.82 ])
  Test R²: 0.716
```

Out[90]:

Target	CV R^2 (mean)	CV R^2 (std)	Test R^2
Type A — Person/Society	0.677	0.064	0.623
Type B — Property	0.753	0.084	0.716

Table 8: RF 5-fold cross-validated R^2 versus held-out test R^2 for both targets. CV R^2 *CV R^2 exceeds test R^2 in both cases, indicating the 80/20 split produced a slightly conservative test set rather than a favourable one. The elevated standard deviation in Type B (σ=0.084) reflects variance across folds — one fold (R^2=0.591) accounts for most of the spread.*

Crime generators by type

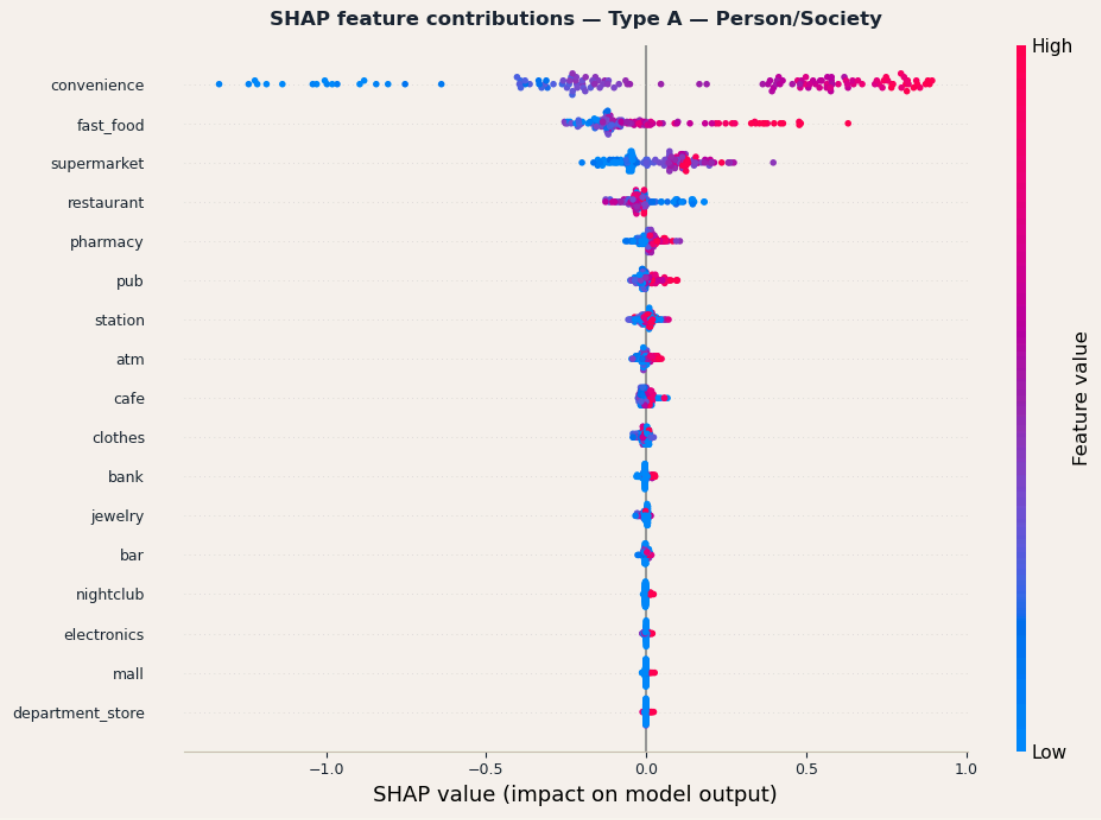
Having established that RF best captures the POI-crime relationship, SHAP values allow us to open the model and ask which amenity types are doing the work — and whether the answer differs by crime type.

Figures 11 and 12 show the SHAP beeswarm plots for Type A and Type B, respectively.

SHAP analysis demonstrates that different amenity categories are primary drivers for each crime type. For Type A (person/society crime), convenience store density is the dominant predictor (maximum SHAP value: 0.75), followed by fast food and ATM density. Notably, traditional night-time economy venues; bars, pubs, and nightclubs, were among the weakest predictors despite their prominent role in criminological discourse. This may reflect the mandatory guardianship associated with licensed premises; door staff, CCTV, and licensing conditions, which suppresses crime opportunity despite high footfall, consistent with the capable guardians component of Routine Activity Theory (Cohen and Felson, 1979).

In [91]:

```
#  SHAP beeswarm plots
import shap

clean_names = [c.replace("_density_log", "") for c in feature_cols]

for target_name, rf in rf_models.items():
    explainer  = shap.TreeExplainer(rf)
    shap_vals  = explainer.shap_values(X_test)

    fig, ax = plt.subplots(figsize=(10, 7), facecolor='#F7F4ED')

    shap.summary_plot(
        shap_vals,
        X_test,
        feature_names=clean_names,
        show=False,
        plot_size=None
    )

    # Style the current figure
    fig = plt.gcf()
    fig.patch.set_facecolor('#F7F4ED')
    ax = plt.gca()
    ax.set_facecolor('#F7F4ED')
    ax.tick_params(colors='#1B2632', labelsize=9)
    ax.spines[['top', 'right']].set_visible(False)
    ax.spines[['left', 'bottom']].set_color('#C9C1B1')

    plt.title(f'SHAP feature contributions — {target_name}',
              fontsize=12, fontweight='bold', color='#1B2632', pad=12)


    plt.tight_layout()
    plt.show()
    plt.close()
```



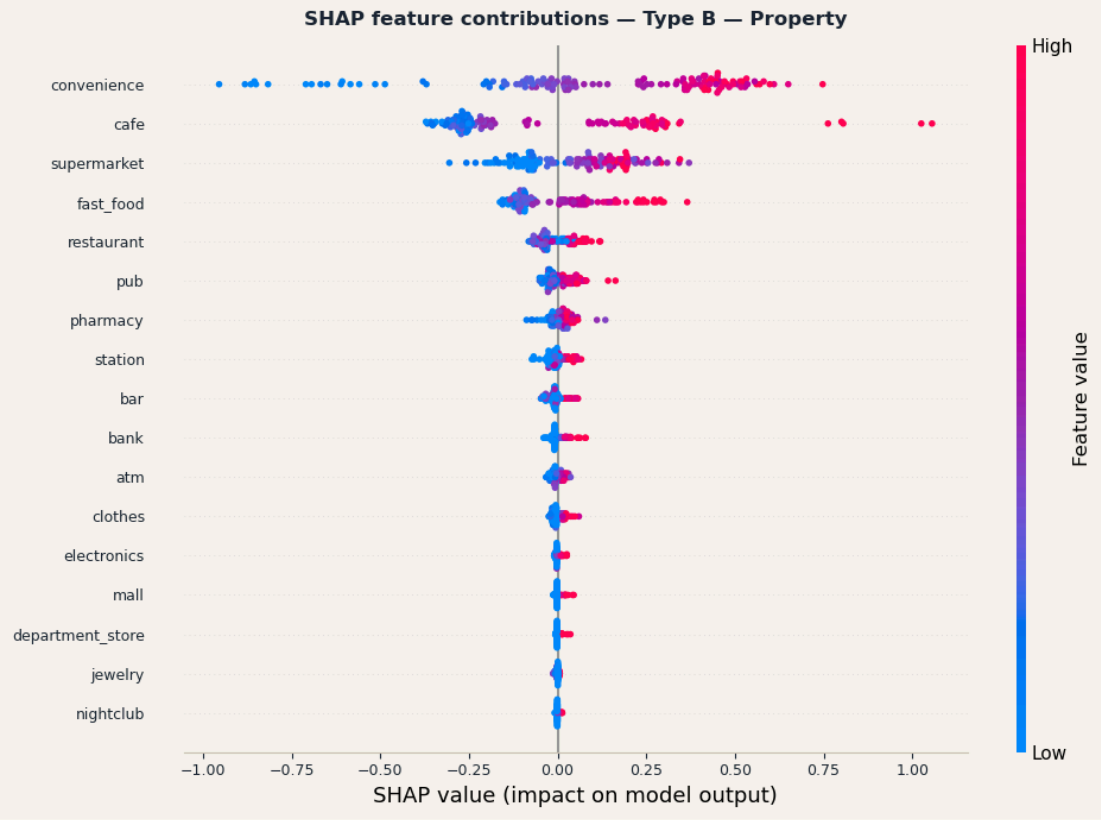


Figure 11: SHAP Feature Contributions — Type A Person and Society Crime *SHAP beeswarm plot for the Random Forest model predicting Type A crime density. Each point represents one ward; colour indicates feature value (red = high, blue = low). Convenience store density is the dominant predictor — high-density wards push predictions strongly positive, while low-density wards extend far left. Fast food and supermarket follow. Night-time economy venues (bars, nightclubs) cluster near zero, confirming their weak predictive contribution.*

> *For Type B (property crime), café density was the primary predictor (maximum SHAP value: 1.25), followed by convenience stores and restaurants. The shift from fast food (Type A) to café (Type B) as the dominant predictor suggests that functionally distinct spatial profiles underlie each crime category. Fast food outlets cluster in high-footfall deprived areas where violence concentrates, while cafés cluster in daytime commercial zones where theft and shoplifting are more prevalent.*

Figure 12: SHAP Feature Contributions — Type B Property Crime *SHAP beeswarm plot for Type B property crime. Convenience store remains the top predictor, but café rises to second — consistent with*

its status as the strongest Pearson correlate of property crime (r=0.84). Café shows a pronounced negative tail, indicating low-café wards are actively suppressed in predicted property crime density. Supermarket and fast food follow.

Spatial diagnostics

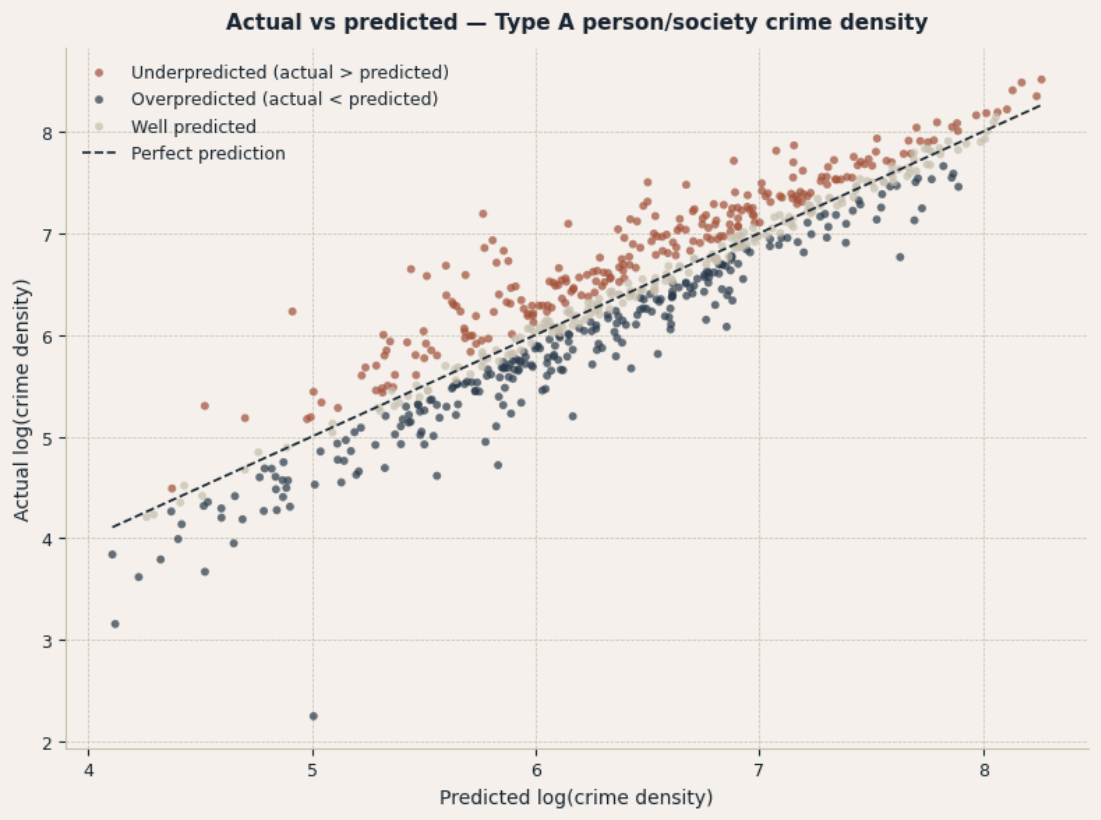
Spatial residuals reveal systematic geographic patterns in model error. Underpredicted wards — where actual crime exceeds model estimates — concentrate in outer east London (Barking and Dagenham, Redbridge) and deprived west London (Brent, Greenwich, Enfield), suggesting socioeconomic deprivation drives crime independently of physical land use. Kensington and Chelsea is a notable exception among underpredicted boroughs — here, underprediction likely reflects high-value tourist targets and opportunity-rich commercial zones rather than deprivation. Overpredicted wards cluster in affluent outer south-west London, where high amenity density does not correspond to elevated crime — likely reflecting the crime-suppressing effects of high property values and private security.

Figures 13 and 14 show the actual vs predicted scatter and spatial residual map for Type A crime.

In [92]:
```
# Interactive actual vs predicted scatter

import plotly.graph_objects as go

# Create pred_df from feature_matrix
pred_df = feature_matrix[["WardCode"] + feature_cols].copy()
pred_df["typeA_pred_log"] = rf_models["Type A — Person/Society"].predict(
    pred_df[feature_cols].fillna(0)
)
pred_df["typeB_pred_log"] = rf_models["Type B — Property"].predict(
    pred_df[feature_cols].fillna(0)
)
pred_df["residual_A"] = feature_matrix["typeA_density_log"].values - pred_df["typeA_pred_log"]
pred_df["residual_B"] = feature_matrix["typeB_density_log"].values - pred_df["typeB_pred_log"]

# Merge ward names
scatter_df = pred_df.merge(
    feature_matrix[["WardCode", "WardName", "Borough"]],
    on="WardCode", how="left"
)
scatter_df["typeA_actual"] = feature_matrix["typeA_density_log"].values
scatter_df["residual_A"]   = scatter_df["typeA_actual"] - scatter_df["typeA_pred_log"]
```

In [93]:
```
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from matplotlib.gridspec import GridSpec

# ── Build scatter_df ──────────────────────────────────────────
scatter_df = pred_df.merge(
    feature_matrix[["WardCode", "WardName", "Borough"]],
    on="WardCode", how="left"
)
scatter_df["typeA_actual"] = feature_matrix["typeA_density_log"].values
scatter_df["residual_A"]   = scatter_df["typeA_actual"] - scatter_df["typeA_pred_log"]

scatter_df["colour"] = scatter_df["residual_A"].apply(
    lambda x: '#A35139' if x > 0.1 else '#2C3B4D' if x < -0.1 else '#C9C1B1'
)
scatter_df["label"] = scatter_df["residual_A"].apply(
    lambda x: 'Underpredicted' if x > 0.1 else 'Overpredicted' if x < -0.1 else 'Well predicted'
)

# ── Figure 13 — Actual vs Predicted scatter ──────────────────
fig, ax = plt.subplots(figsize=(8, 6), facecolor='#F7F4ED')
ax.set_facecolor('#F7F4ED')

for group, color, name in [
    ('Underpredicted', '#A35139', 'Underpredicted (actual > predicted)'),
    ('Overpredicted',  '#2C3B4D', 'Overpredicted (actual < predicted)'),
    ('Well predicted', '#C9C1B1', 'Well predicted'),
]:
    subset = scatter_df[scatter_df['label'] == group]
    ax.scatter(
        subset['typeA_pred_log'],
        subset['typeA_actual'],
        c=color, s=18, alpha=0.7,
        label=name, edgecolors='none'
    )

mn = scatter_df['typeA_pred_log'].min()
mx = scatter_df['typeA_pred_log'].max()
ax.plot([mn, mx], [mn, mx],
        color='#1B2632', linewidth=1.2,
        linestyle='--', label='Perfect prediction')

ax.set_xlabel('Predicted log(crime density)', fontsize=10, color='#1B2632')
ax.set_ylabel('Actual log(crime density)',    fontsize=10, color='#1B2632')
ax.set_title('Actual vs predicted — Type A person/society crime density',
             fontsize=11, fontweight='bold', color='#1B2632', pad=10)
```

```python
ax.tick_params(colors='#1B2632', labelsize=9)
ax.spines[['top', 'right']].set_visible(False)
ax.spines[['left', 'bottom']].set_color('#C9C1B1')
ax.xaxis.grid(True, color='#C9C1B1', linewidth=0.5, linestyle='--')
ax.yaxis.grid(True, color='#C9C1B1', linewidth=0.5, linestyle='--')
ax.set_axisbelow(True)
ax.legend(fontsize=9, framealpha=0, labelcolor='#1B2632')

plt.tight_layout()
plt.show()

# ── Summary table ──────────────────────────────────────────────
summary = (
    scatter_df.groupby("label")
    .agg(
        Wards        = ("WardCode",   "count"),
        Mean_Residual= ("residual_A", "mean"),
        Max_Abs      = ("residual_A", lambda x: x.abs().max())
    )
    .round(3)
    .reset_index()
    .rename(columns={
        "label":         "Prediction Group",
        "Mean_Residual": "Mean Residual",
        "Max_Abs":       "Max |Residual|"
    })
    .sort_values("Wards", ascending=False)
)
summary["% of Wards"] = (
    summary["Wards"] / summary["Wards"].sum() * 100
).round(1).astype(str) + "%"

row_colors = {
    'Underpredicted': '#F0D5CC',
    'Overpredicted':  '#C9D4DE',
    'Well predicted': '#E8E4DC',
}

fig2, ax2 = plt.subplots(figsize=(8, 2), facecolor='#F7F4ED')
ax2.set_facecolor('#F7F4ED')
ax2.axis('off')

cols = list(summary.columns)
cell_text = summary.values.tolist()
cell_colors = [
    [row_colors.get(row[0], '#F7F4ED')] * len(cols)
    for row in cell_text
]

table = ax2.table(
    cellText=cell_text,
    colLabels=cols,
    cellLoc='center',
    loc='center',
    cellColours=cell_colors
)
table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1, 1.6)

for (row, col), cell in table.get_celld().items():
    cell.set_edgecolor('#C9C1B1')
    cell.set_linewidth(0.5)
    if row == 0:
        cell.set_facecolor('#2C3B4D')
        cell.get_text().set_color('#EEE9DF')
        cell.get_text().set_fontweight('bold')
    else:
        cell.get_text().set_color('#1B2632')

plt.tight_layout()
plt.show()
```



Prediction Group	Wards	Mean Residual	Max |Residual|	% of Wards
Overpredicted	252	-0.311	2.759	37.1%
Underpredicted	238	0.357	1.426	35.1%
Well predicted	189	-0.009	0.099	27.8%

Figure 13: Actual vs Predicted — Type A Person and Society Crime Density *Scatter plot of actual against predicted log crime density for all 679 wards, coloured by residual group. Overall fit is strong — 27.8% of wards are well predicted (|residual| ≤ 0.1); 37.1% overpredicted and 35.1% underpredicted. Overpredicted wards (blue, mean residual −0.311) cluster at lower predicted values, suggesting the model overestimates crime in low-density outer wards. Underpredicted wards (rust, mean residual +0.357) are more evenly distributed, indicating crime driven by factors beyond POI density.*

In [94]:
```
worst_over = scatter_df.nsmallest(1, "residual_A")[["WardName", "Borough", "residual_A"]]
print(worst_over)
```

```
   WardName  Borough  residual_A
577   Darwin  Bromley   -2.759068
```

In [95]:
```
# ── Build map_gdf ──────────────────────────────────────────────
map_gdf = ward_gdf[["WardCode", "WardName", "Borough", "geometry"]].copy()
map_gdf = map_gdf.merge(
    feature_matrix[["WardCode", "typeA_density_log"]],
    on="WardCode", how="left"
)
map_gdf["typeA_pred_log"] = rf_models["Type A — Person/Society"].predict(
    feature_matrix[feature_cols].fillna(0)
)
map_gdf["residual_A"] = (
    map_gdf["typeA_density_log"] - map_gdf["typeA_pred_log"]
).round(3)

# Remove Aviation Security artefact
map_gdf = map_gdf[
    ~map_gdf["Borough"].str.contains('Aviation|SO18', case=False, na=False)
].copy()

map_gdf = map_gdf.to_crs("EPSG:27700")
gla_plot = gla_gdf.to_crs("EPSG:27700")

# Colormap and norm
from matplotlib.colors import TwoSlopeNorm, LinearSegmentedColormap
div_cmap = LinearSegmentedColormap.from_list(
    'residual', ['#2C3B4D', '#F7F4ED', '#A35139']
)
vmax = map_gdf["residual_A"].abs().quantile(0.95)
norm = TwoSlopeNorm(vmin=-vmax, vcenter=0, vmax=vmax)
```

In [96]:

```python
import geopandas as gpd

# ── Borough boundaries ──────────────────────────────────────
# Dissolve wards to borough level
borough_gdf = map_gdf.dissolve(by='Borough').reset_index()

# Get top underpredicted boroughs by mean residual
top_under = (
    map_gdf.groupby('Borough')['residual_A']
    .mean()
    .sort_values(ascending=False)
    .head(5)
    .index.tolist()
)

top_over = (
    map_gdf.groupby('Borough')['residual_A']
    .mean()
    .sort_values(ascending=True)
    .head(5)
    .index.tolist()
)

# ── Plot ────────────────────────────────────────────────────
fig, ax = plt.subplots(figsize=(10, 8), facecolor='#F7F4ED')
ax.set_facecolor('#F7F4ED')

gla_plot.plot(ax=ax, color='#D8D3C8', edgecolor='none')

map_gdf.plot(
    column='residual_A',
    cmap=div_cmap,
    norm=norm,
    legend=True,
    linewidth=0.1,
    edgecolor='#C9C1B1',
    legend_kwds={
        'shrink':  0.6,
        'aspect':  25,
        'pad':     0.02,
        'format':  '%.1f',
        'label':   'Residual'
    },
    ax=ax,
    missing_kwds={'color': '#C9C1B1'}
)

gla_plot.boundary.plot(ax=ax, color='#1B2632', linewidth=1.0)

# ── Highlight top underpredicted boroughs ──────────────────
under_gdf = borough_gdf[borough_gdf['Borough'].isin(top_under)]
over_gdf  = borough_gdf[borough_gdf['Borough'].isin(top_over)]

under_gdf.boundary.plot(
    ax=ax, color='#A35139', linewidth=1.8, linestyle='--'
)
over_gdf.boundary.plot(
    ax=ax, color='#2C3B4D', linewidth=1.8, linestyle='--'
)

# ── Borough centroid labels ────────────────────────────────
for _, row in under_gdf.iterrows():
    cx, cy = row.geometry.centroid.x, row.geometry.centroid.y
    ax.annotate(
        row['Borough'],
        xy=(cx, cy),
        fontsize=7, color='#A35139',
        fontweight='bold', ha='center', va='center',
        bbox=dict(boxstyle='round,pad=0.2',
                  facecolor='#F7F4ED',
                  edgecolor='none', alpha=0.7)
    )

for _, row in over_gdf.iterrows():
    cx, cy = row.geometry.centroid.x, row.geometry.centroid.y
    ax.annotate(
        row['Borough'],
        xy=(cx, cy),
        fontsize=7, color='#2C3B4D',
        fontweight='bold', ha='center', va='center',
        bbox=dict(boxstyle='round,pad=0.2',
                  facecolor='#F7F4ED',
                  edgecolor='none', alpha=0.7)
    )

# ── Legend for highlights ──────────────────────────────────
from matplotlib.lines import Line2D
legend_elements = [
    Line2D([0], [0], color='#A35139', linewidth=1.8,
           linestyle='--', label='Top 5 underpredicted boroughs'),
    Line2D([0], [0], color='#2C3B4D', linewidth=1.8,
           linestyle='--', label='Top 5 overpredicted boroughs'),
]
```

```python
ax.legend(handles=legend_elements, fontsize=8,
          framealpha=0.8, labelcolor='#1B2632',
          loc='lower left', facecolor='#F7F4ED',
          edgecolor='#C9C1B1')

# ── Bounds ──────────────────────────────────────────────
minx, miny, maxx, maxy = gla_plot.total_bounds
pad_x = (maxx - minx) * 0.02
pad_y = (maxy - miny) * 0.02
ax.set_xlim(minx - pad_x, maxx + pad_x)
ax.set_ylim(miny - pad_y, maxy + pad_y)

ax.set_title('Spatial residuals — Type A person/society crime',
             fontsize=11, fontweight='bold', color='#1B2632', pad=10)
ax.text(0.5, 1.005,
        'Rust = underpredicted · Blue = overpredicted · '
        'Dashed outlines = top 5 boroughs by mean residual',
        transform=ax.transAxes, ha='center',
        fontsize=8, color='#615D56', style='italic')
ax.set_axis_off()

plt.tight_layout()
plt.show()
```

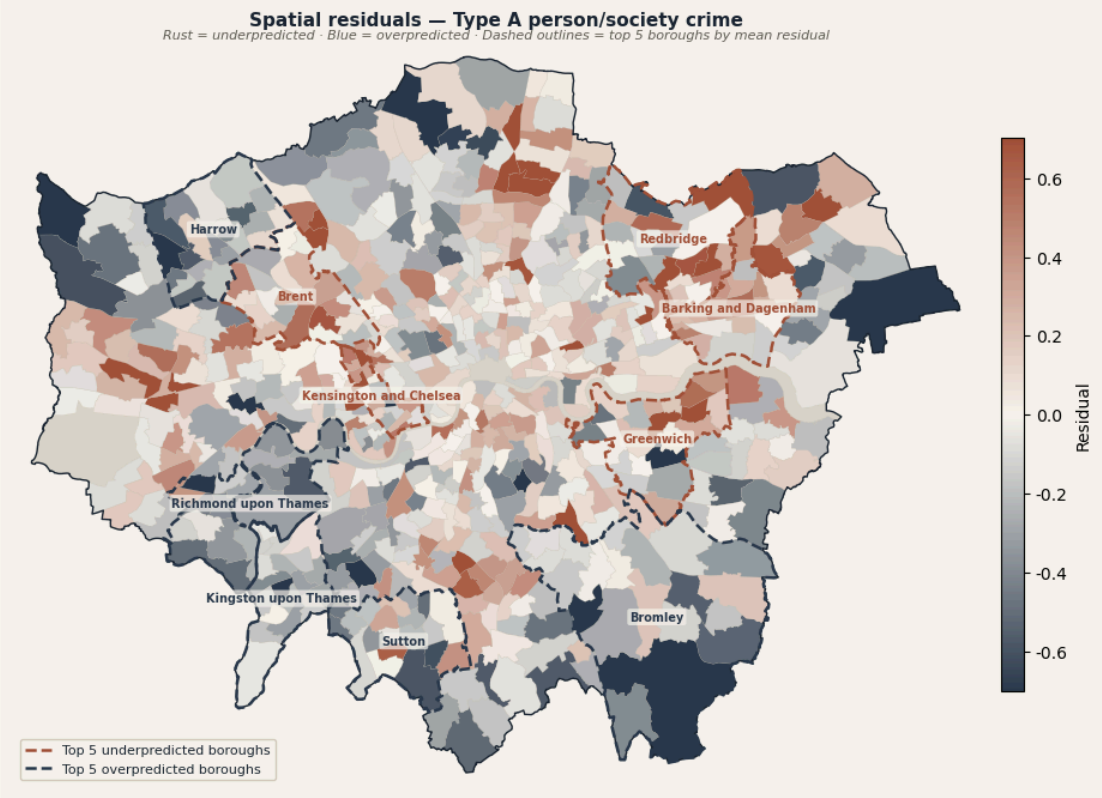


Figure 14: Spatial Residuals — Type A Person and Society Crime *Choropleth map of Random Forest residuals across 679 London wards. Rust indicates underprediction; blue indicates overprediction. Dashed outlines mark the top five boroughs by mean residual in each direction. Underprediction concentrates in outer east London (Barking and Dagenham, Redbridge) and deprived west London (Brent), consistent with socioeconomic deprivation driving crime beyond what POI density predicts. Kensington and Chelsea also underpredicts — likely reflecting high-value tourist targets rather than deprivation. Overprediction clusters in affluent outer south-west London, where high amenity density does not translate to elevated crime.*

These findings are summarised below, along with limitations and directions for future work.

Limitations

1. **Dark figure of crime** — Police-recorded data undercounts actual crime, systematically more so in deprived areas, likely compounding underprediction in those boroughs.
2. **Ecological fallacy** — Ward-level aggregation means findings describe area-level patterns only and cannot be attributed to individuals.
3. **OSM coverage** — POI data is uneven across London, with sparser coverage in outer boroughs potentially underrepresenting amenity density in those areas.
4. **Absence of controls** — No temporal, demographic, or socioeconomic variables were included, limiting causal inference.

5. **Boundary mismatch** — 25 wards in the 2024 ONS boundary revision had no matching MPS crime records, reducing the analytical sample from 704 to 679 wards.

Conclusion

[go back to the top]

This study demonstrates that open-source POI data alone can predict ward-level crime density with moderate to strong accuracy, without recourse to demographic or socioeconomic variables. The key finding is that amenity type matters as much as amenity density. Convenience stores and fast food are the primary generators of person and society crime; cafés dominate property crime prediction. Most strikingly, night-time economy venues — bars, pubs, and nightclubs — showed unexpectedly weak predictive power despite their prominence in criminological literature. This is not a null result: it suggests that the regulatory environment of licensed premises actively suppresses crime opportunity, and that sustained daytime footfall is a stronger generator of crime in London wards than night-time activity. This has direct policy implications — amenity density alone is an insufficient proxy for crime risk.

Where the model fails systematically — underpredicting in deprived outer boroughs, overpredicting in affluent ones; socioeconomic factors appear to operate independently of physical land use. POI density captures the physical opportunity structure of crime; socioeconomic variables capture the social conditions that generate it. These frameworks are complementary, and future work combining both has the potential to provide a more complete account of crime determinants across London.

In [97]:
```
elapsed = time.time() - start_time
print(f"Runtime: {elapsed/3600:.2f} hours ({elapsed/60:.1f} minutes)")
```

```
Runtime: 0.01 hours (0.8 minutes)
```

References

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